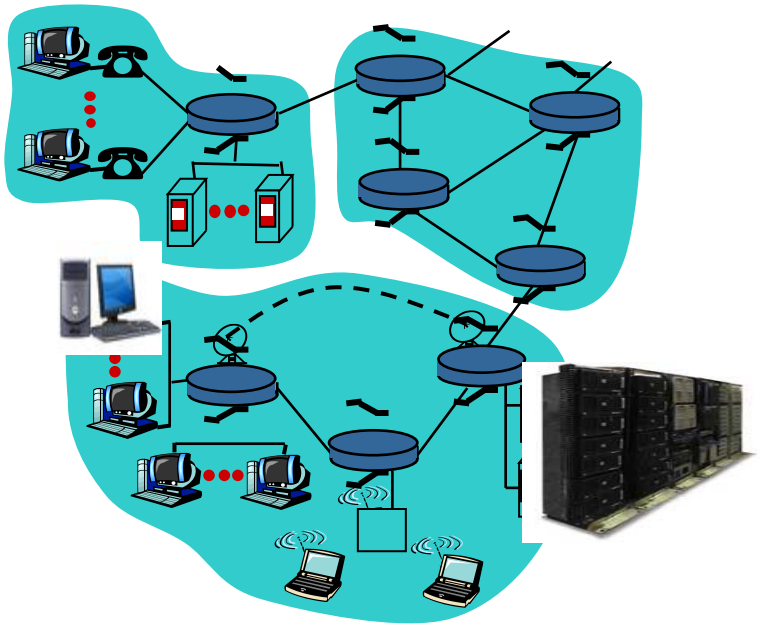
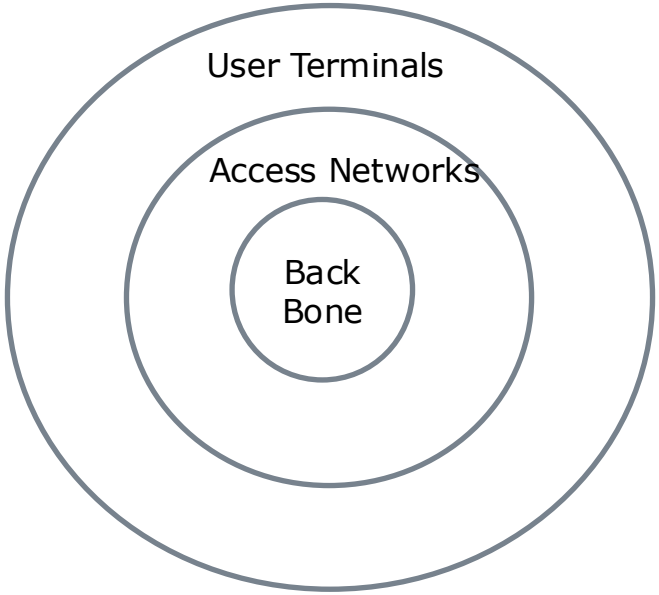
Tecnologie di Interconnessione: IoT

Objectives

- Focus on communication technologies for the IoT
- Energy is an issue
- A bit of comm technologies
  - Cables: field bus
  - No cables: Long range (LoRaWAN), Short Range: (ZigBee vs THREAD vs BT)

The “Classical” Internet

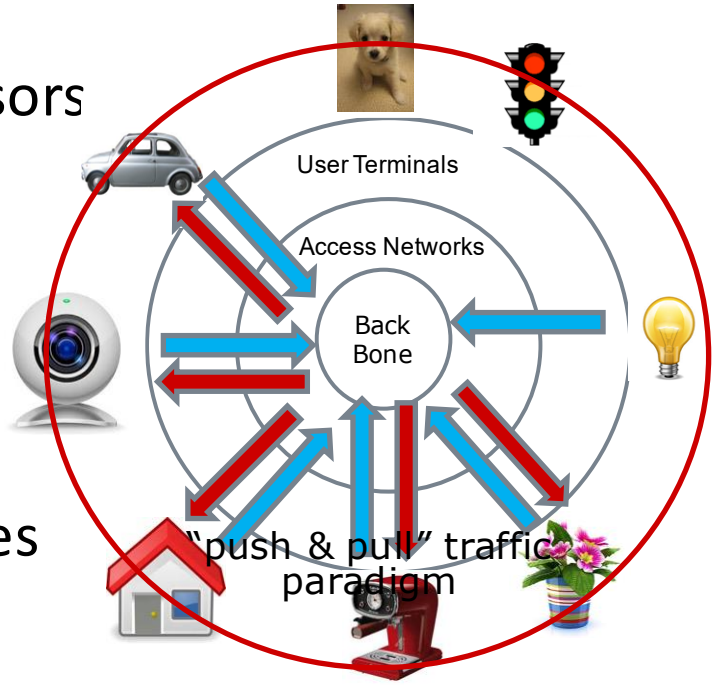




- Mostly Static Clients accessing “Well Known” services in the network

IoT Technology Leaps

- Devices
  - Size: CMOS miniaturization, micro-sensors (MEMS, materials and circuits)
  - Energy: fuel cells batteries, energy scavenging
  - Mobility
- Comm. Protocols
  - Low power wireless comm. technologies
- Cloud-based resources
  - Diffusion of cloud services, architectures and processing power
- AI
  - Deep Learning, Generative AI

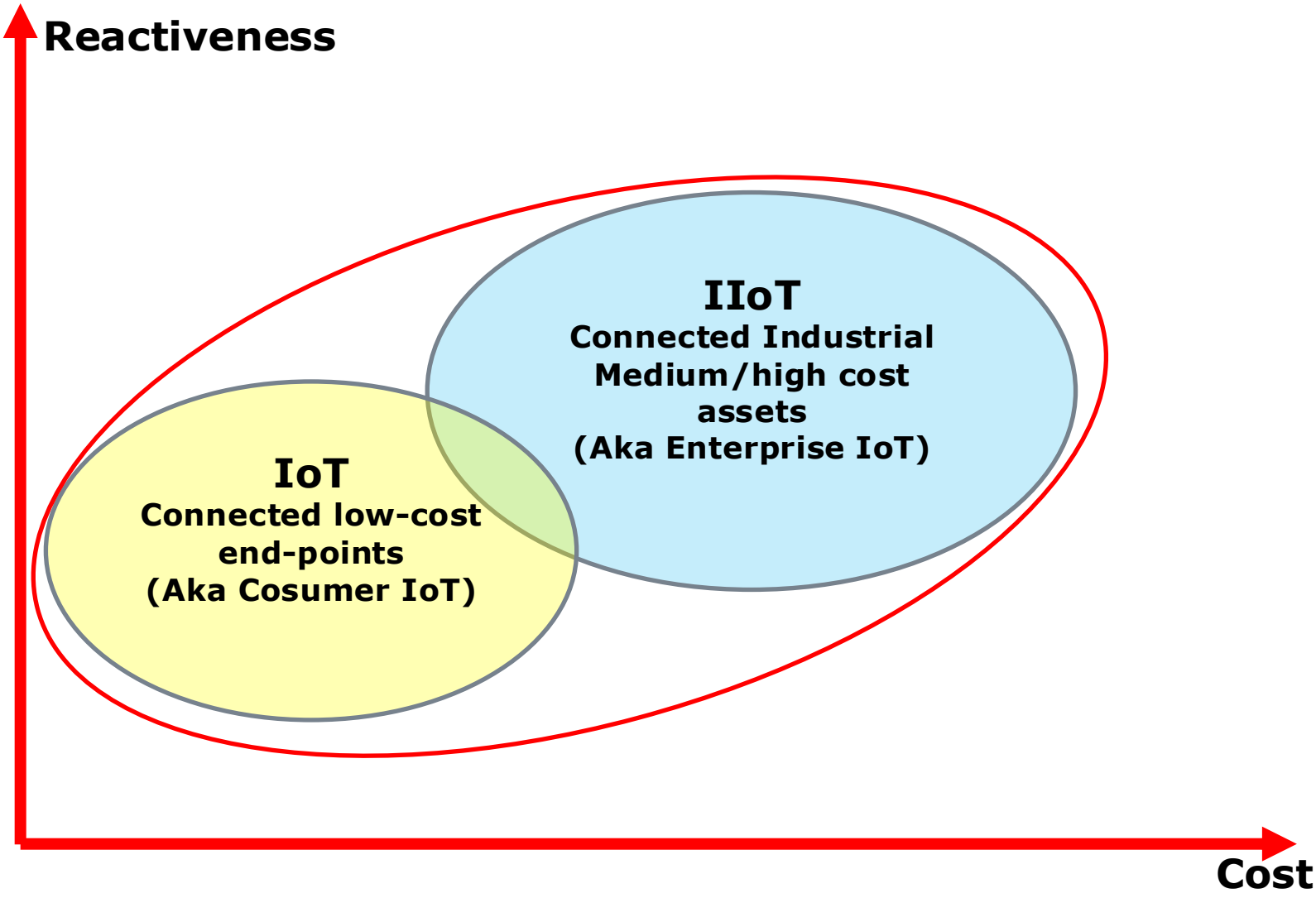


Ambient Intelligence – Science Fiction

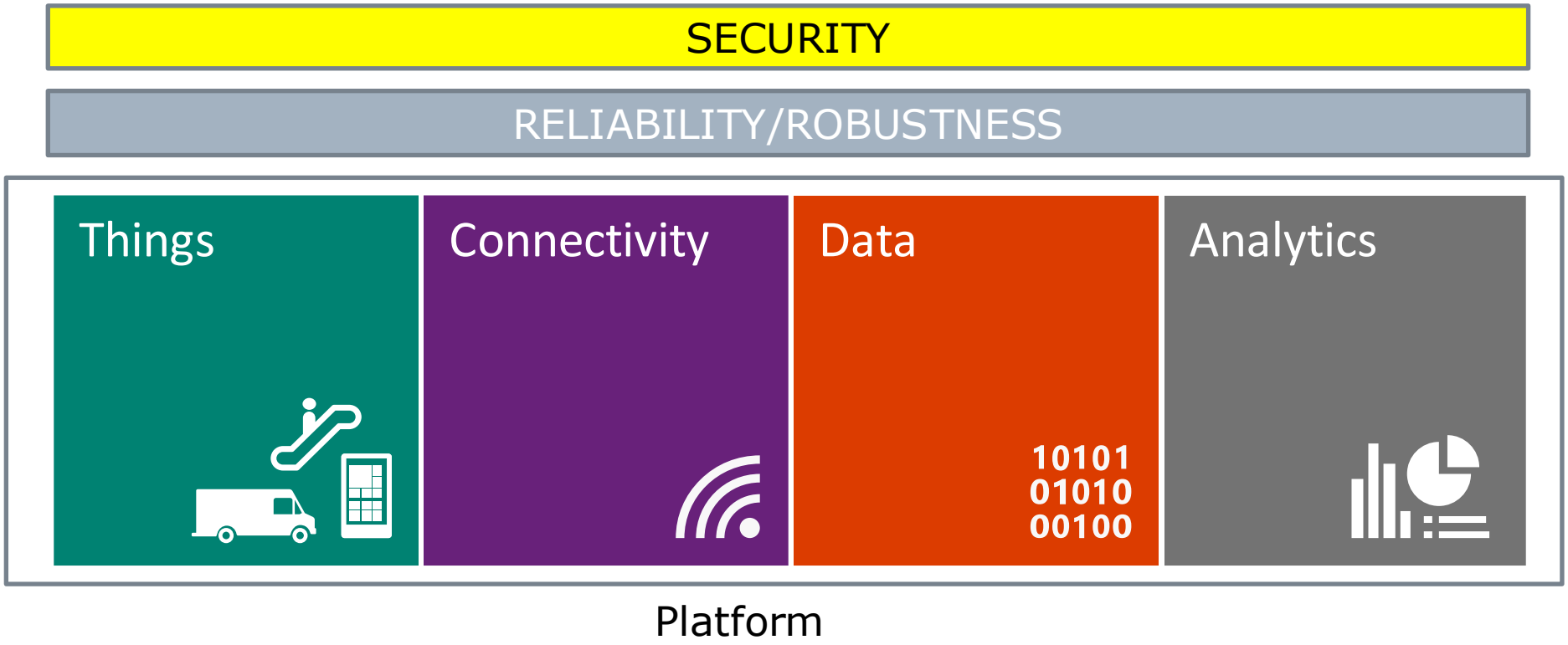


Minority Report, 2002 Steven Spielberg

IIoT and IoT


Reactiveness
IIoT
Connected Industrial
Medium/high cost
assets
(Aka Enterprise IoT)
IoT
Connected low-cost
end-points
(Aka Cosumer IoT)
Cost

IoT Building Blocks and Challenges



Field Devices for IoT

Hardware and Hardware Abstractions

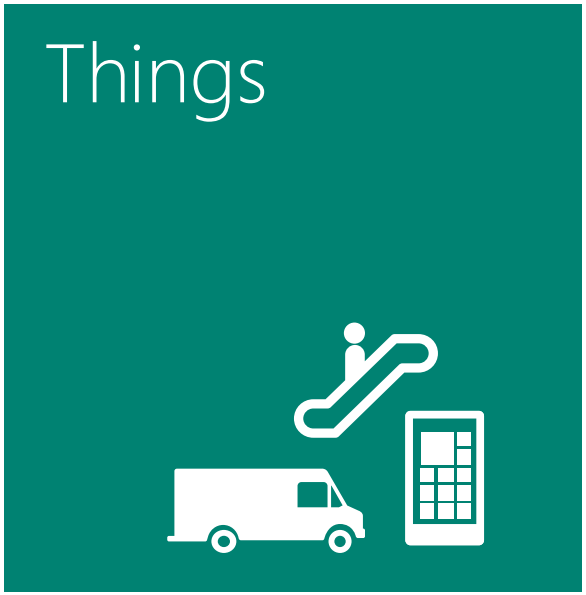
Design Challenges/Guidelines/Opportunities

Let's start From the edge..

- A Sensor Node (or *mote*) is a device with the following capabilities:
  - Sensing external phenomena
  - Processing information
  - Storing information
  - Communicating with other motes or devices
- The Actuator's tasks
  - Receiving input signals from control devices
  - Processing/storing information
  - Acting on the industrial process

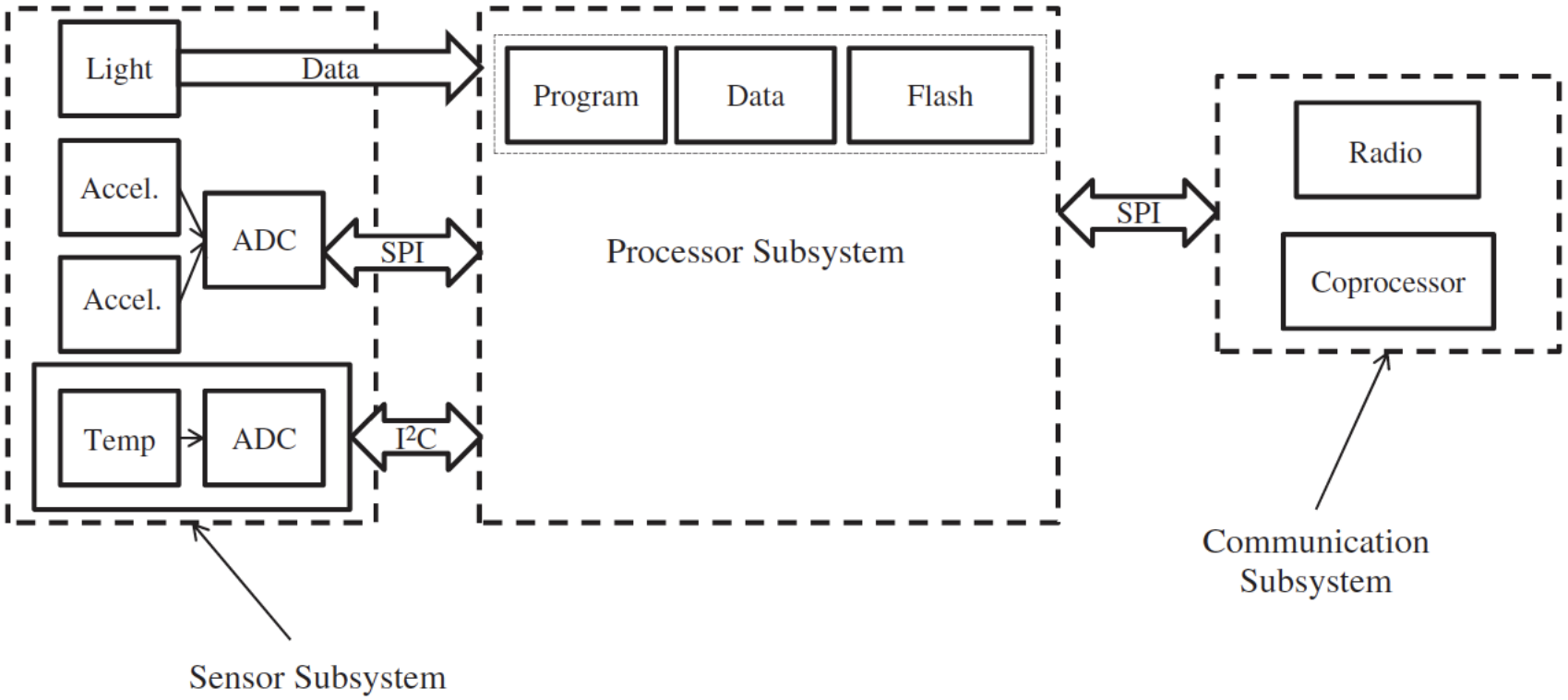
Any examplels??

The “Things”



- IT/OT check list:
  - What do you need to sense?
  - What constraint you have on:
    - Wired/Wireless
    - Flexibility/Programmability
    - Integration/size/design
    - Budget
    - H&M interfaces

Mote Architecture



Examples

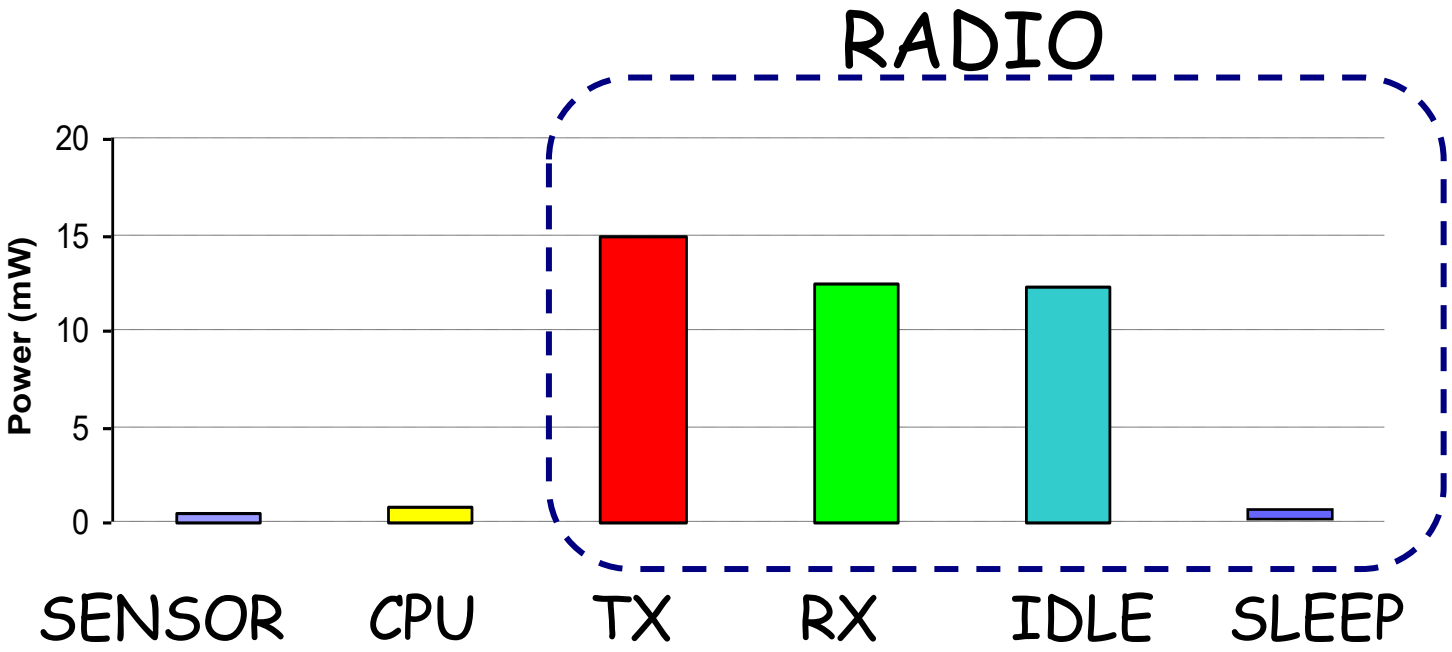
	COM	CPU	BATTERY	STORAGE	SENSORS
	GPS, cellular, WiFi, BLE	S5, 64 bit dual core	Li-Po 296 mAh	32GB	Optical heart, accelerometer, gyroscope, ambient light, microphone, speaker, gymkit
	WiFi, BLE	32-bit ARM Cortex M3	Li-Ion 560 mAh	1MB Flash, 128 kB RAM	Accelerometer, Gyroscope, MFS, Light, Temperature, Pressure, Humidity

Energy efficiency is “THE” Issue

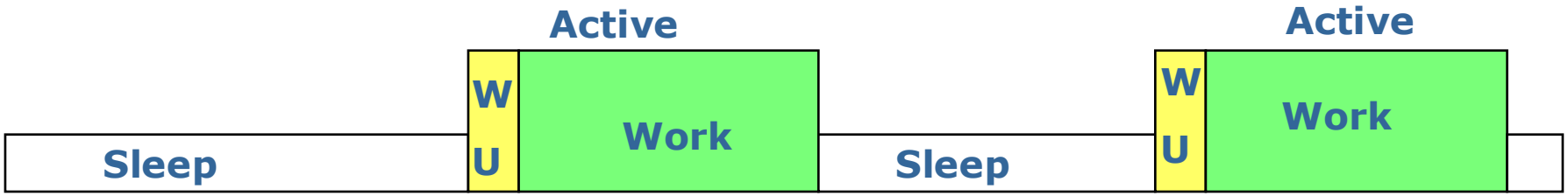
- Sensor node LIFETIME depends on BATTERY lifetime
- Recharging and/or battery replacement may be immaterial or too expensive

Goal: Provide as much energy as possible at smallest cost/volume/weight/recharge

Source: Lecture slides of “Sensor Networks”, Prof. I. Akyildiz



Operation Mode: Duty-Cycling



- Average power
  - $P_{ave} = = (1-f_{active})*P_{sleep} + f_{active}*P_{active}$
  - $P_{ave} = f_{sleep}*P_{sleep} + f_{wakeup}*P_{wakeup} + f_{work}*P_{work}$
- Lifetime
  - EnergyStore / (P_{ave} - P_{gen})
- f_{work} is the duty cycle (typical values < 1%)

Energy Consumption for Communication

TRANSMISSION



$$E_{tx} = P_{tx}(T_{wu} + T_{tx}) + P_o T_{tx}$$

RECEPTION



$$E_{rx} = P_{rx}(T_{wu} + T_{rx})$$

where

P_{tx} is power consumed by transmitter

P_{rx} is power consumed by receiver

P_O is output power of transmitter

T_{tx} time to transmit a packet

T_{rx} time to receive a packet

T_{wu} is start-up time for transmitter

On the emitted power

- The emitted power is often a tunable parameter
- Good practice is to set it to the lowest value which allows for “good reception”
- The quality of the reception process is “measured” in terms of
  - Bit Error Rate (BER): fraction of bit not correctly received (“1” for a “0” or viceversa)
  - Packet Error Rate (PER): fraction of packet not correctly received
  - PER/BER relation (packet of length *l*, independent errors):

$$PER = 1 - (1 - BER)^l$$

Signal to noise and Interference Ratio

- BER (and PER in turn) depends on the "level of noise" in the TX/RX channel, which, in turn, depends on the transmitted/received power

$$\text{SINR} = 10 \log_{10} \left(\frac{P_{\text{recv}}}{N_0 + \sum_{i=1}^{k} I_i} \right)$$

Thermal noise: *KTB*

- BER can be computed once given the specific TX/RX channel (modulation) and the specific SINR

Receiver Sensitivity

- Each receiver is characterized by a *sensitivity* parameter (e.g. P_{min}=-95dBm),

  The minimum input signal power needed at receiver input to provide adequate SNR at receiver output to do data demodulation

- Example: IEEE 802.15.4
  - Receiver sensitivity (packet error rate < 1%)
    - P_{min}>−85 dBm @ 2.4 GHz band
    - P_{min}>−92 dBm @ 868/915 MHz band
- Knowing such parameter, one can find the required emitted power at the transmitter by inverting the propagation law of the channel to get to required emitted power

Emitted power - Example

- A wireless receiver is characterized by a sensitivity P_r=0.1[uW]; the transmitter is d=10[m] from the receiver; the TX-RX is performed at a carrier frequency f=2.4GHz; the propagation on the channel is characterized by the following model

$$P_r = P_t g_t g_r (\frac{\lambda}{4\pi d})^2$$ *

- Further assuming the antenna gains equal to 1, we get a required emitted power

$$P_t = P_r (\frac{4\pi d}{\lambda})^2 \approx 100[mW]$$

*If you have no clue on where this model comes from you may want to refresh wireless propagation notions

Communication Technologies for the IoT: introduction

Connectivity in I4.0 – Why is it so different?

Connectivity

- IT/OT Check list:
  - What is the network size
  - Mobile or Fixed Nodes
  - Any service QoS constraint (throughput, latency)?
  - The “integration issue”
  - Any budget?



What is the traffic in IoT?

- ***Cyclic Traffic***: periodic transfer of sensor measurements and set points

  Example: velocity control loop of a electrical engine requires sending a set point (2 bytes) every 100us

- ***Acyclic Traffic***: traffic generated by unpredictable events

  Example: alarm triggered by a failing component of the industrial process

- ***Multimedia Traffic***: images/video

  Example: intrusion detection systems, defect inspection

- ***Backhand Traffic:*** *aggregated flows*

Networking KPIs

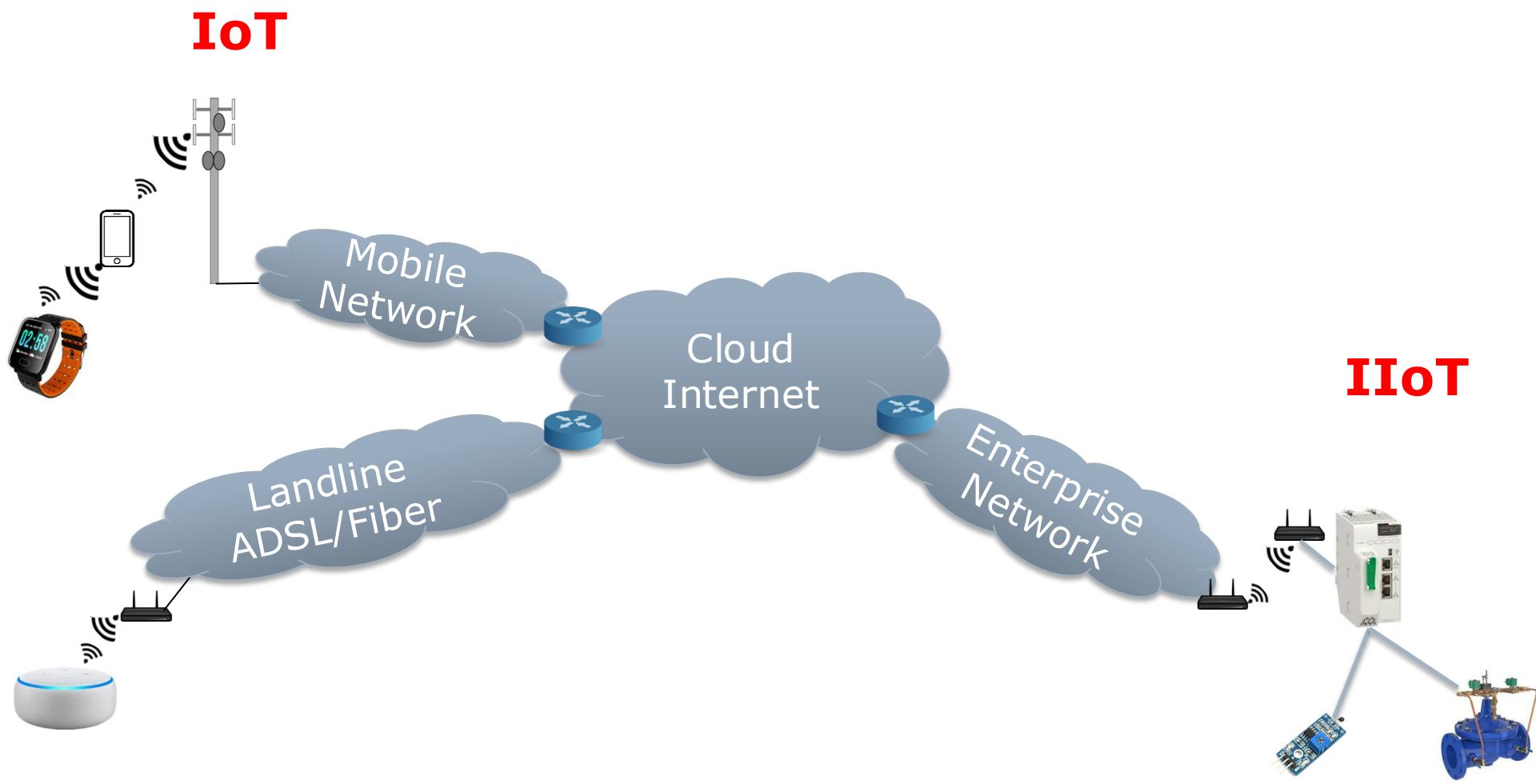
- *Reliability*
  - *Bit Error Rate (BER):* *bit error probability*
  - *Packet Error Rate (PER):* *packet error probability*
- *Speed*
  - *Throughput (rate)*: number of bits (bytes) per second that can be sent across a link (Unit measure: [bit/s] or [byte/s]); depends on link type
- *Latency*
  - *Packet Delivery Time*: time needed to send one *service data unit* from source to destination (Unit measure: [s]); includes transmission and propagation time
  - *Protocol Time:* time needed to execute the communication protocol
  - *Minimum Cycle Time (MCT):* minimum time required to execute a cycle in a control loop
  - *Jitter:* precision/reliability in performing periodic operations

$$J = \frac{|P - \hat{P}|}{P}$$

$\hat{P}$: *actual task execution time*

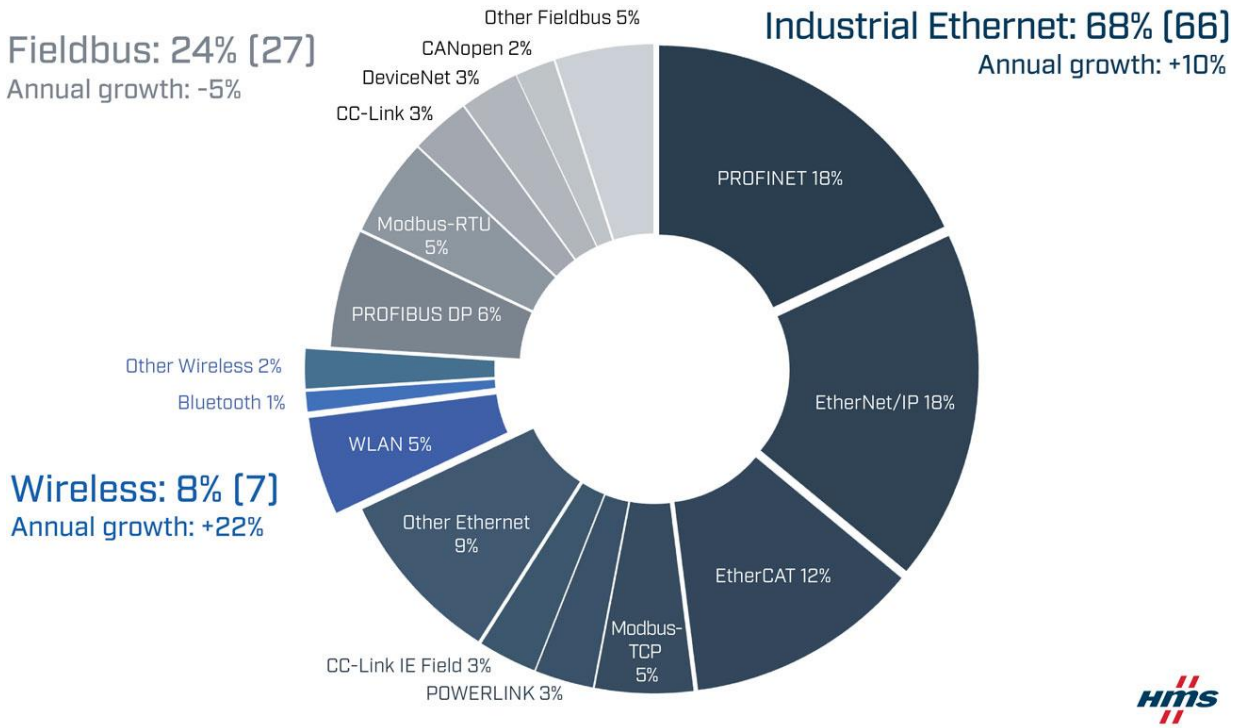
P: nominal task execution time

IoT Connectivity: Never (seldom) one single-hop/technology



Communication technologies for the IoT: wired technologies

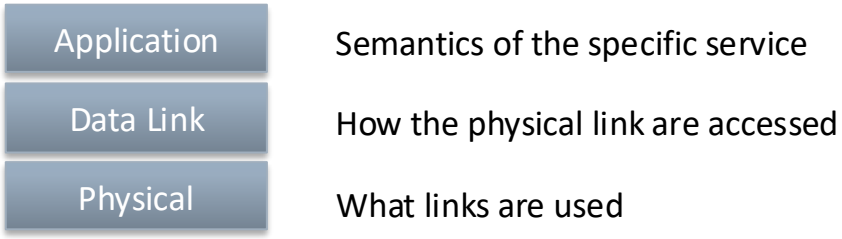
Industrial Communication Technologies – Market Share



https://www.hms-networks.com/.it

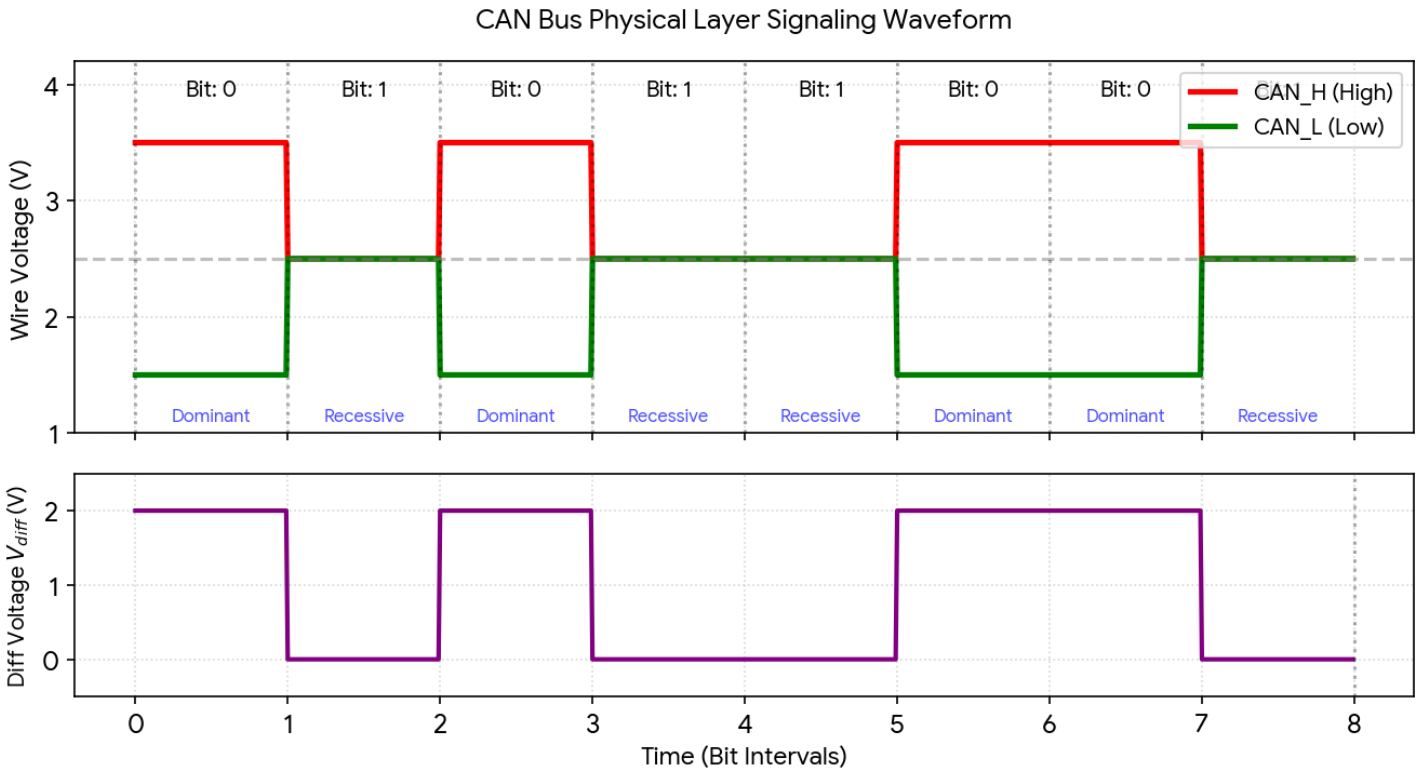
Fieldbus Generalities

- Mainly targeting the interconnection of control devices and field devices (sensors/actuators)
- Short frequently-exchanged messages
- Tigth requirements in terms of delay and determinism



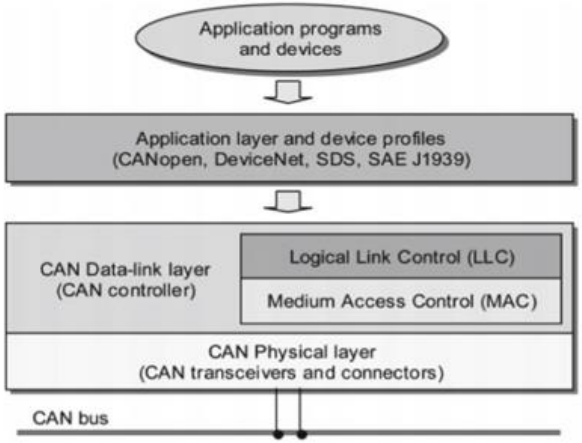
Fieldbus Example: the CAN Bus

- Shared Physical Bus implemented through a twisted pair (CAN High, CAN Low)
- NRZ Differential Voltage Modulation



Rate: 1 [Mb/s]

Fieldbus Example: The CAN bus



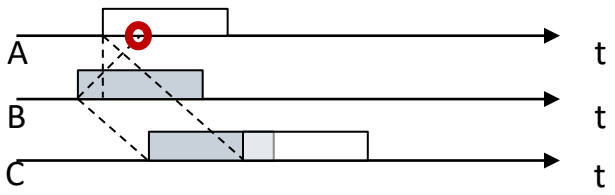


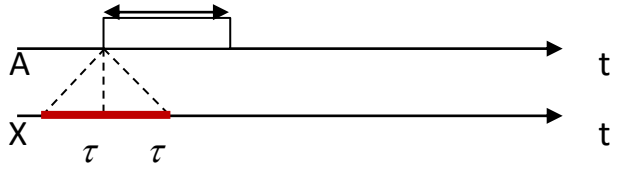
CSMA to arbitrate collisions

- Carrier Sensing Multiple Access

sense the BUS before transmitting; if BUS free transmit otherwise refrain and try later

- Collisions

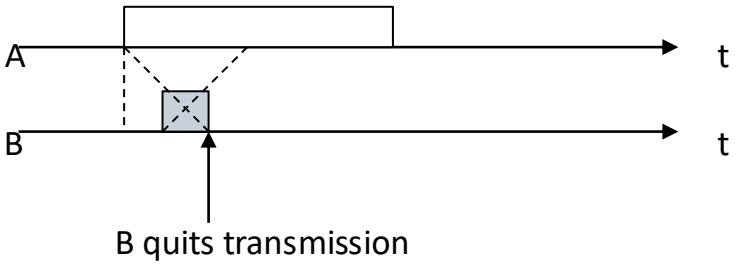




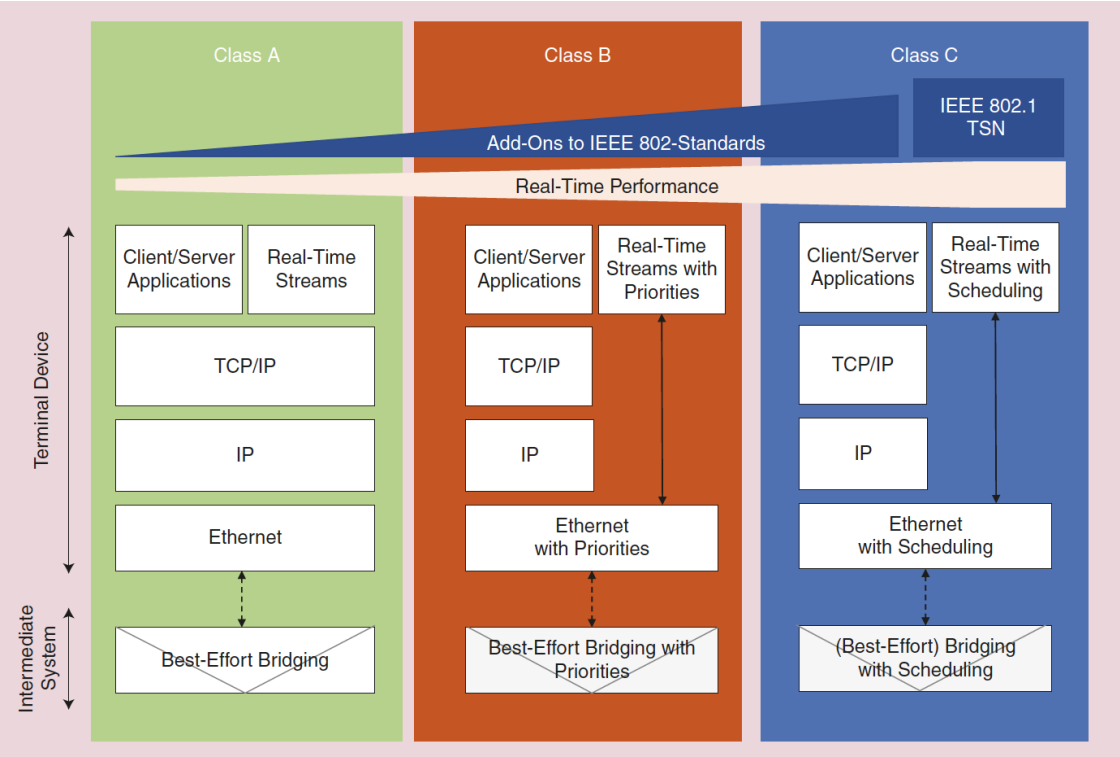
Vulnerability period

BUS Arbitration

- Each message has a priority (Lower identifier field means higher priority)
- Each station monitors its own transmission and the status of the BUS
- If a transmitting station overhears another transmission on the channel at higher priority, then quits (e.g., B in the figure)



Ethernet @ Industry 4.0



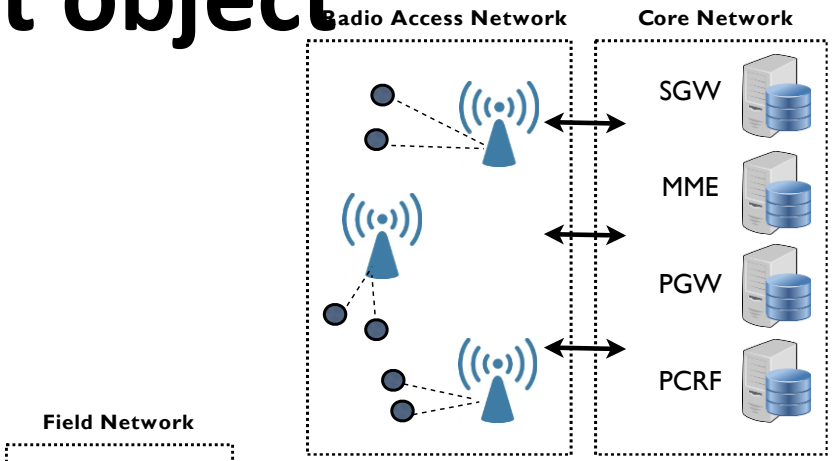
- Class A: MCT around 100[ms]
  - Ethernet/IP
  - Foundation Fieldbus
- Class B: MCT around 10[ms]
  - Profinet
- Class C: MCT around 1[ms]
  - EtherCAT
  - Profinet IRT
  - Ethernet Powerlink

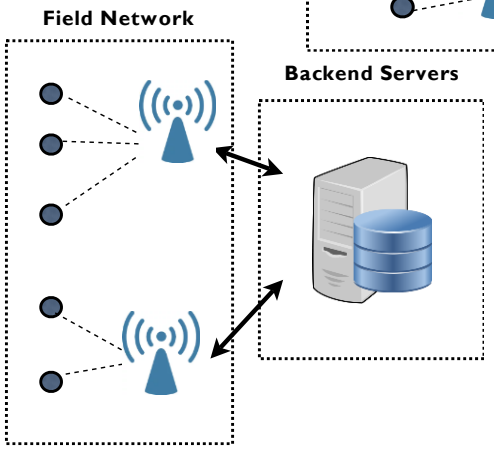
M. Wollschlaeger, T. Sauter, J. Jasper, *The Future of Industrial Communications*, IEEE Industrial Electronics Magazine, 2017

Communication technologies for the IoT: wireless technologies

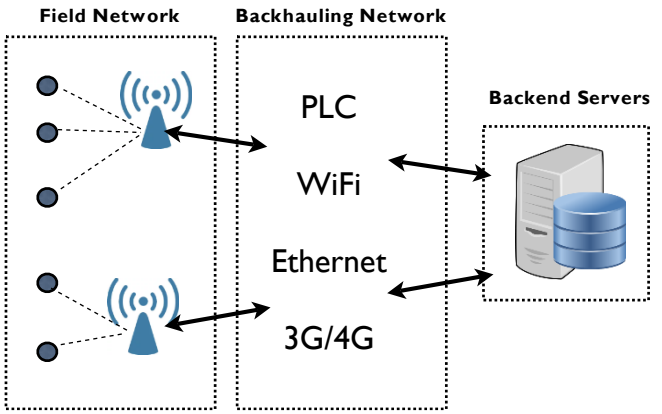
The Race to the Smart object

- Long range
  - Mobile Radio Networks
    - RAN and CN Evolutions
  - Cellular IoT Operators
    - Low Power Long Range Technology

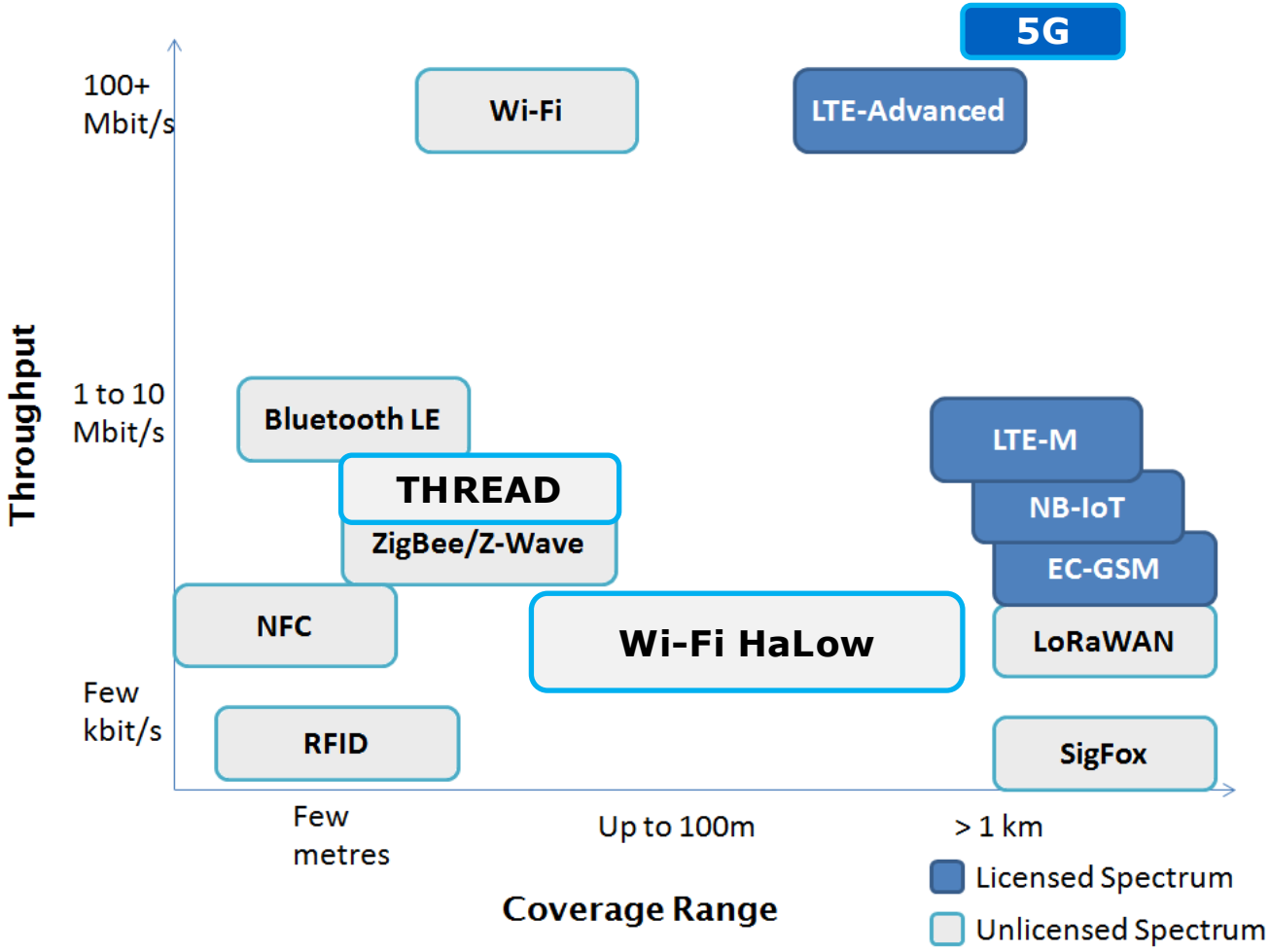




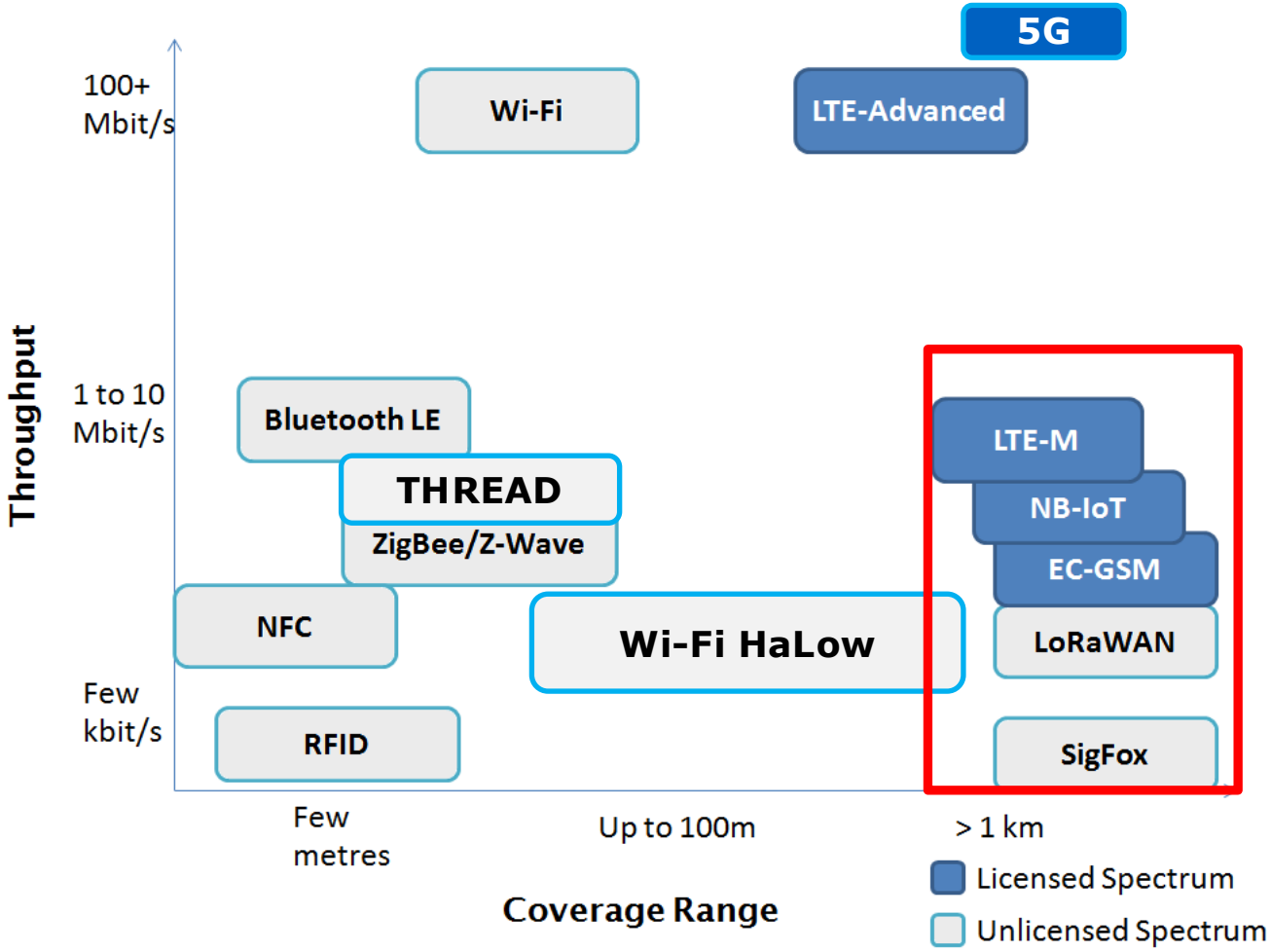
- Short range
  - Short/medium range + backhauling



Connectivity Offer



Lang Range Communication Technologies



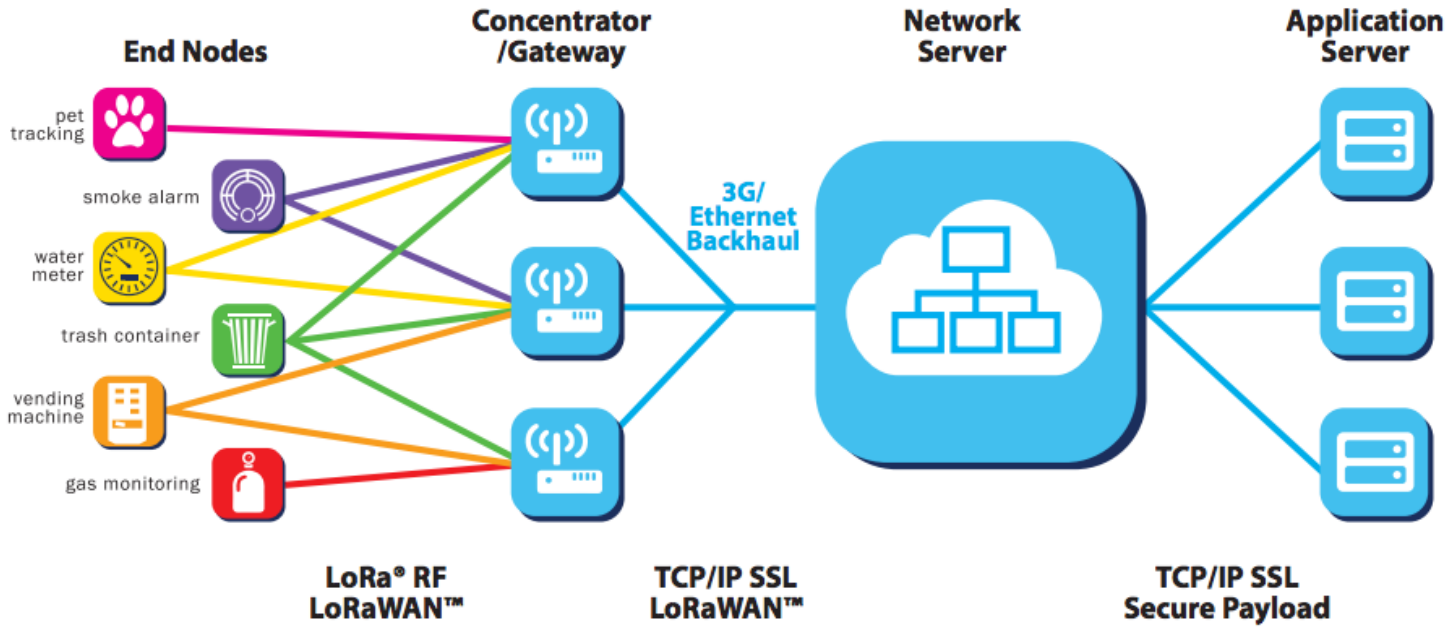
Cellular IoT Operators

- Value proposition
  - Lowest TCO
  - Lowest Energy Consumption
  - Out of the box connectivity
  - Global Reach

	SigFOX	LoRaWAN		Weightless			Ingenu
		EU	US	-W	-N	-P	
Spectrum	868-902MHz	863-870MHz 433MHz	902-928MHz	470-790MHz TV white spaces	sub GHz (ISM)	sub GHz (ISM)	2450 MHz
Channel Width	100Hz	125kHz-250kHz	125kHz-500kHz	6-8MHz	200Hz	12.5kHz	1MHz
TX Rate UL	≤100b/s	250b/s-50 b/s	980b/s-kb/s	250b/s-50kb/s	250b/s	200b/s -100kb/s	624kb/s
TX Rate DL	256b/day	250b/s-50 kb/s	980b/s-21.9kb/s	2.5kb/s-16Mb/s	none	200bytes -100kb/s	156[kb/s]
Packet Size	≤ 12bytes	≤ 222bytes	≤ 222 bytes	≥ 10 bytes	≤ 20bytes	≥ 10 bytes	6bytes-10kbytes
Max Range	10-50km	2-15km		5km	3km	2km	100km
TX power UL	10μW-100mW	14dBm	20dBm	17dBm	17dBm	17dBm	20dBm
Standard (if any)	Proprietary	standard available	standard available	standard available	standard available	standard in the works	proprietary

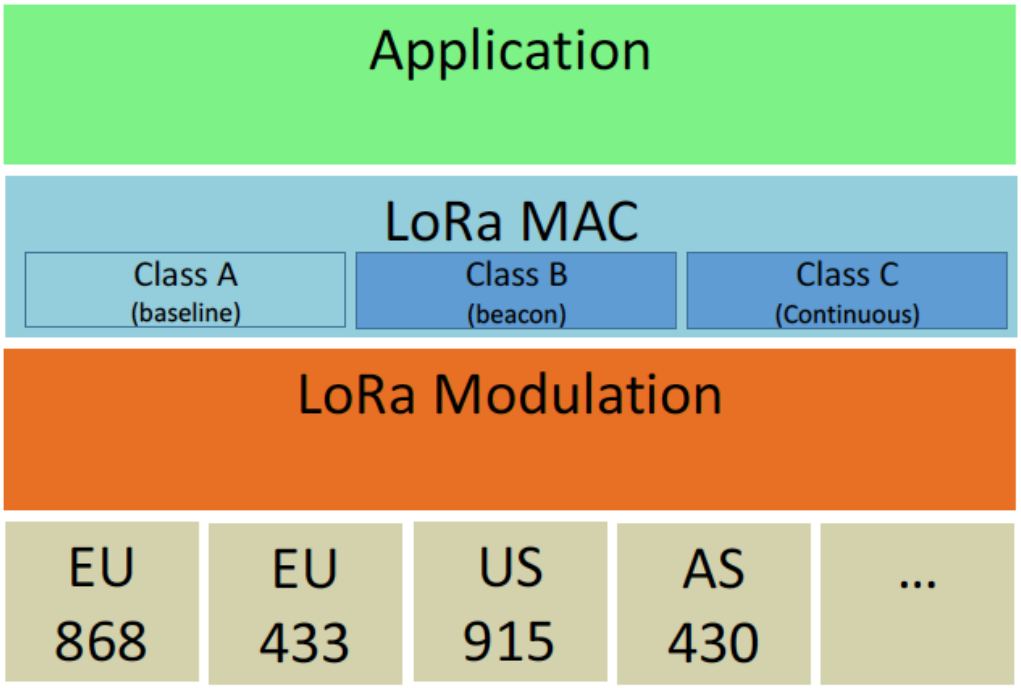
LoraWAN Architecture

A Technical Overview of LoRa and LoraWAN, LoraAlliance



- Association-less, cellular-like architecture
- End devices: field devices
- Gateways: receive and forward messages from end devices (and network server)
- Network server: where all the intelligence is
  - Remove duplicate messages, manages ACKs, manages radio link parameters, etc.

LoraWAN Protocol Stack



Open source

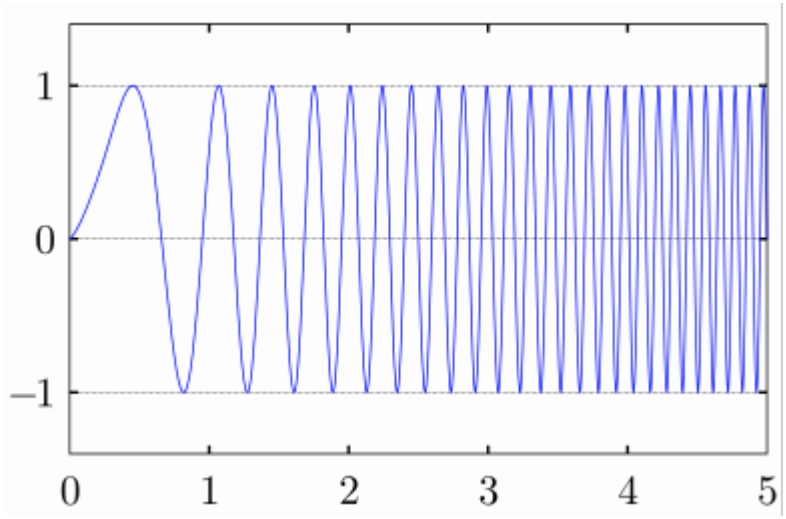
Proprietary
By SemTech™

Region-specific
Carrier frequency [MHz]

LoRa™ Modulation

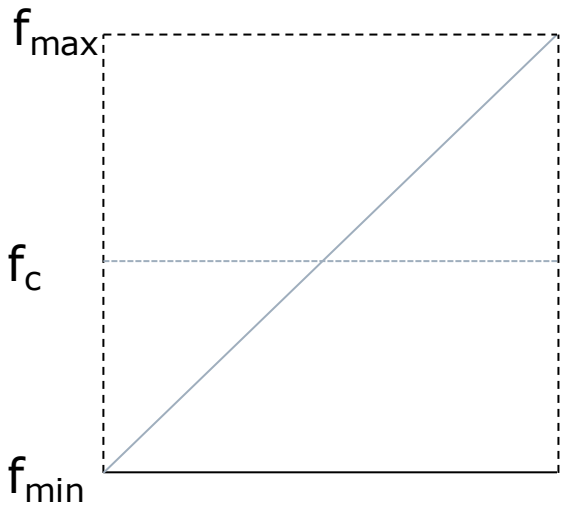
- Proprietary chirp-based spread spectrum
- Signal is chipped at a higher data rate and modulated onto a chirp signal

amplitude



time

frequency



time

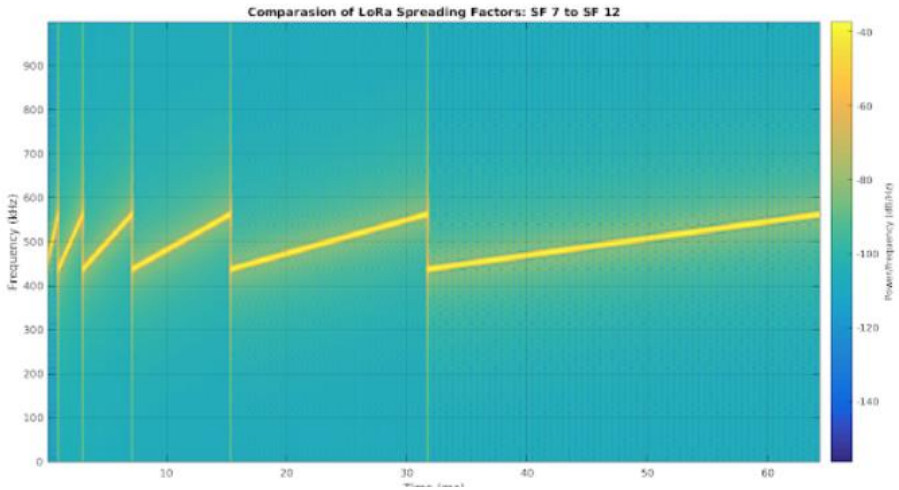
Chip and bit Rates

- The chip rate [chip/s] and the nominal bit rate [bit/s] are related by the following formulas

$$R_c = BW \quad [chips/s]$$

$$R_b = SF \frac{\frac{4}{4+CR}}{\frac{2^{SF}}{BW}} \quad [b/s]$$

- Where BW is the reference bandwidth and SF is the adopted *spreading factor*



How to Choose?

- Two contrasting objectives:
  - Rate: “I want to go fast” (low SF)
  - Reliability: “I want to go safe” (high SF)

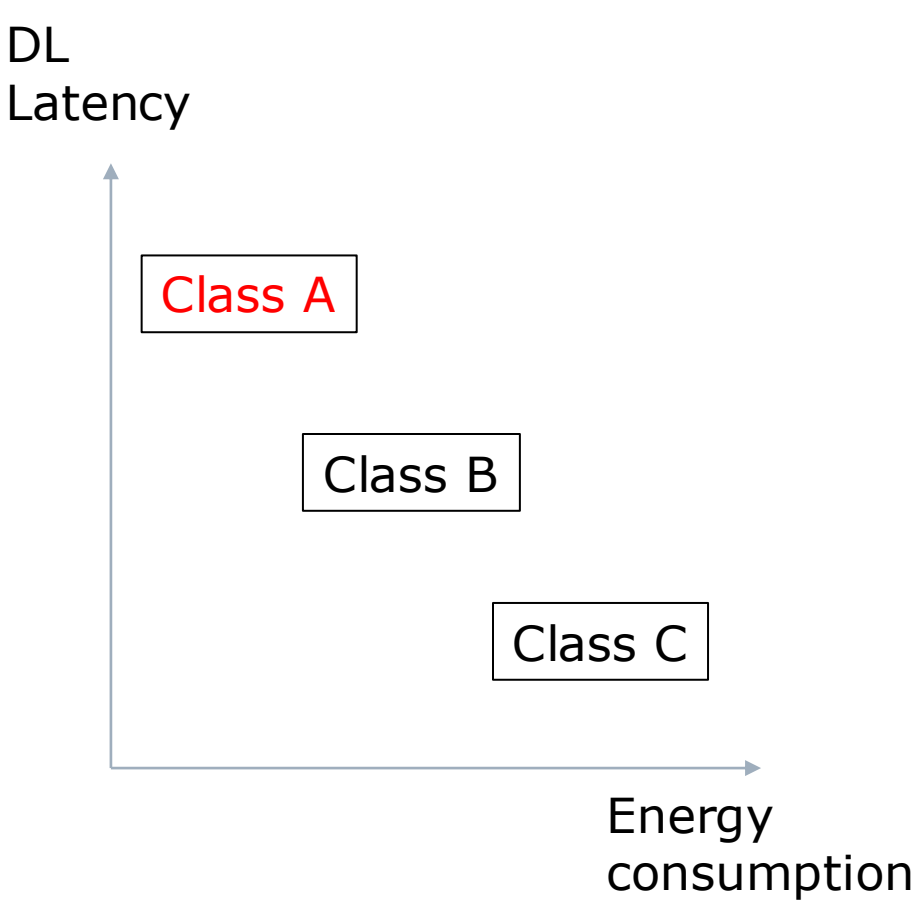
DataRate	Configuration	Indicative physical bit rate [bit/s]
0	LoRa: SF12 / 125 kHz	250
1	LoRa: SF11 / 125 kHz	440
2	LoRa: SF10 / 125 kHz	980
3	LoRa: SF9 / 125 kHz	1760
4	LoRa: SF8 / 125 kHz	3125
5	LoRa: SF7 / 125 kHz	5470
6	LoRa: SF7 / 250 kHz	11000
7	FSK: 50 kbps	50000
8..15	RFU	

M. Bor, et al., Do LoRa Low-Power Wide-Area Networks Scale?, Mswim 2016

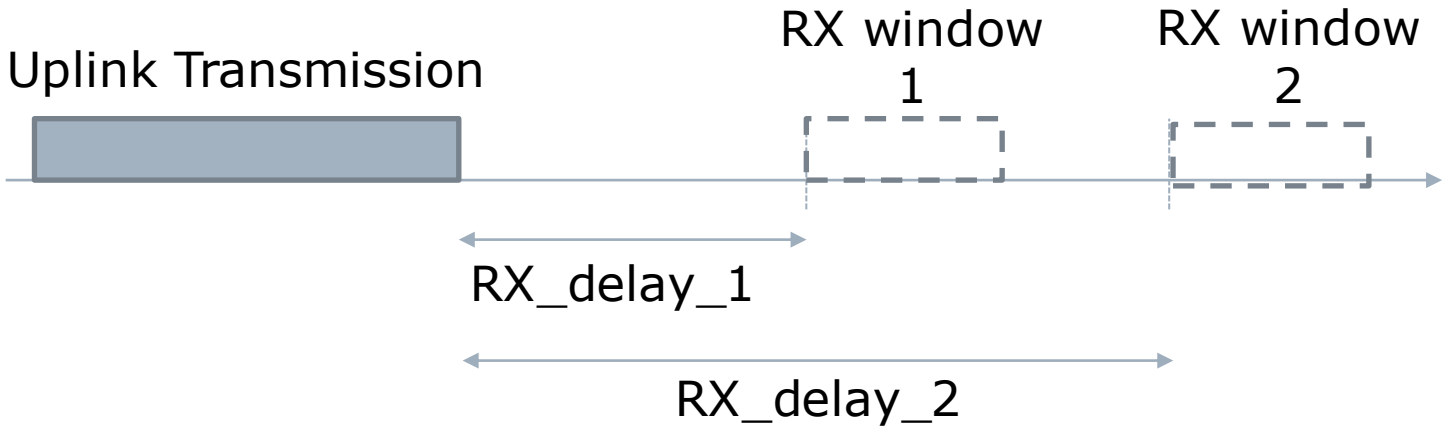
	Bandwidth (kHz)		
SF	125	250	500
7	−126.50	−124.25	−120.75
8	−127.25	−126.75	−124.00
9	−131.25	−128.25	−127.50
10	−132.75	−130.25	−128.75
11	−134.50	−132.75	−128.75
12	−133.25	−132.25	−132.25

LoraWAN End Devices

- *Class A:*
  - Downlink “slots” following uplink transmissions
  - ALOHA-like access in the uplink
- *Class B:*
  - “extra” downlink ”slots” available in a scheduled fashion
- *Class C:*
  - Almost continuous downlink “slots”



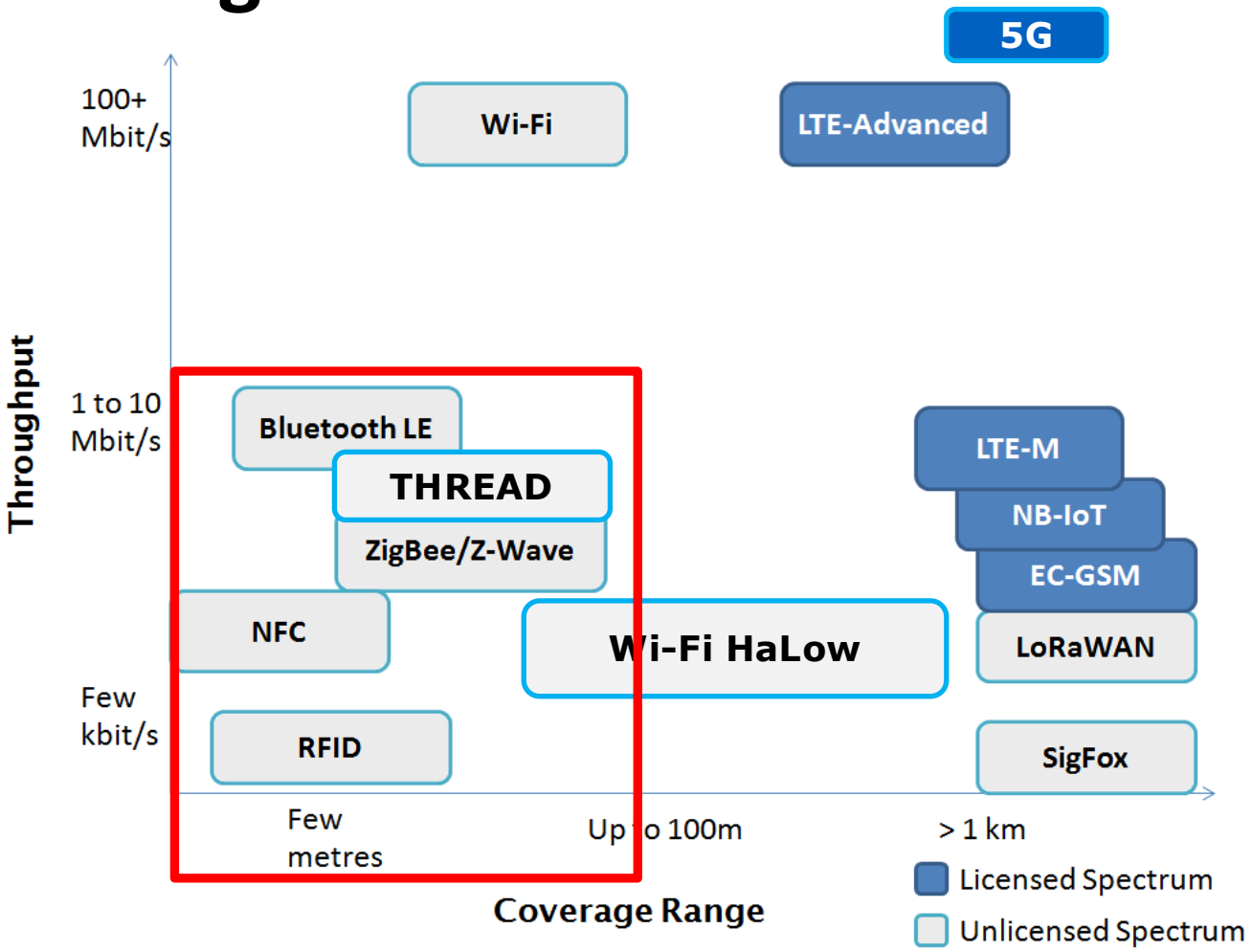
LoraWAN Class A Devices



- In EU: RX_delay_1 = 1[s]; RX_delay_2= RX_delay_1 + 1[s]
- If a preamble is detected during one of the receive windows, the radio receiver stays active until the downlink frame is demodulated
- If a frame was detected and subsequently demodulated during the first receive window and the frame was intended for this end-device after address and MIC (message integrity code) checks, the end-device does not open the second receive window

Short Range Communication Technologies and Protocols

Short Range Communication Technologies



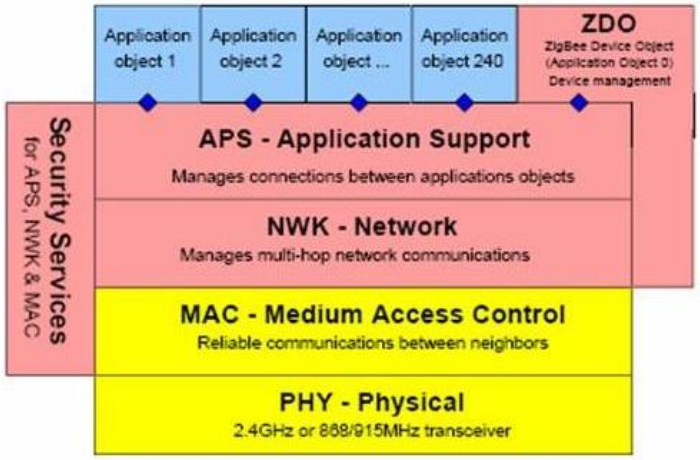
- Highly fragmented technologies/standards/protocols
- Fine for small-scale, hot spot coverage
- The “Gateway Problem”

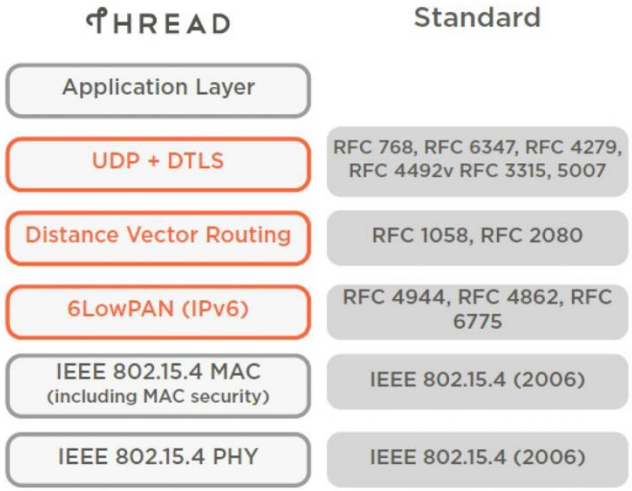
Several Solutions available

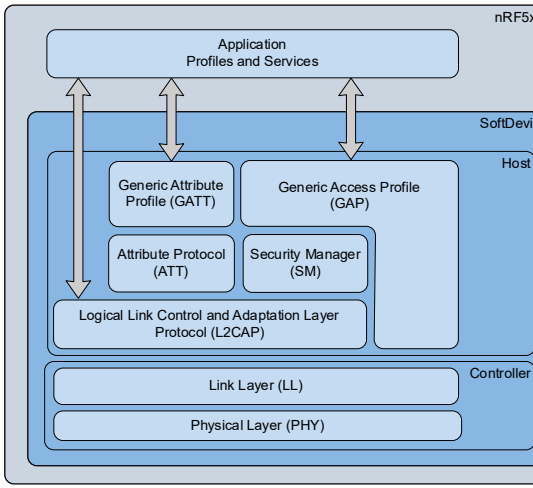
- Classification Guidelines
  - **Proprietary** (WirelessHART) vs **open standard** (WiFi, ZigBee, 6LowPAN, THREAD)
  - **Application specific** vs **application agnostic**
    - ZWAVE for home automation, WirelessHART for industrial applications; 6LowPAN/THREAD for everything
  - **IP-compliant** vs **non-IP-compliant**

ZigBee vs THREAD vs BLE

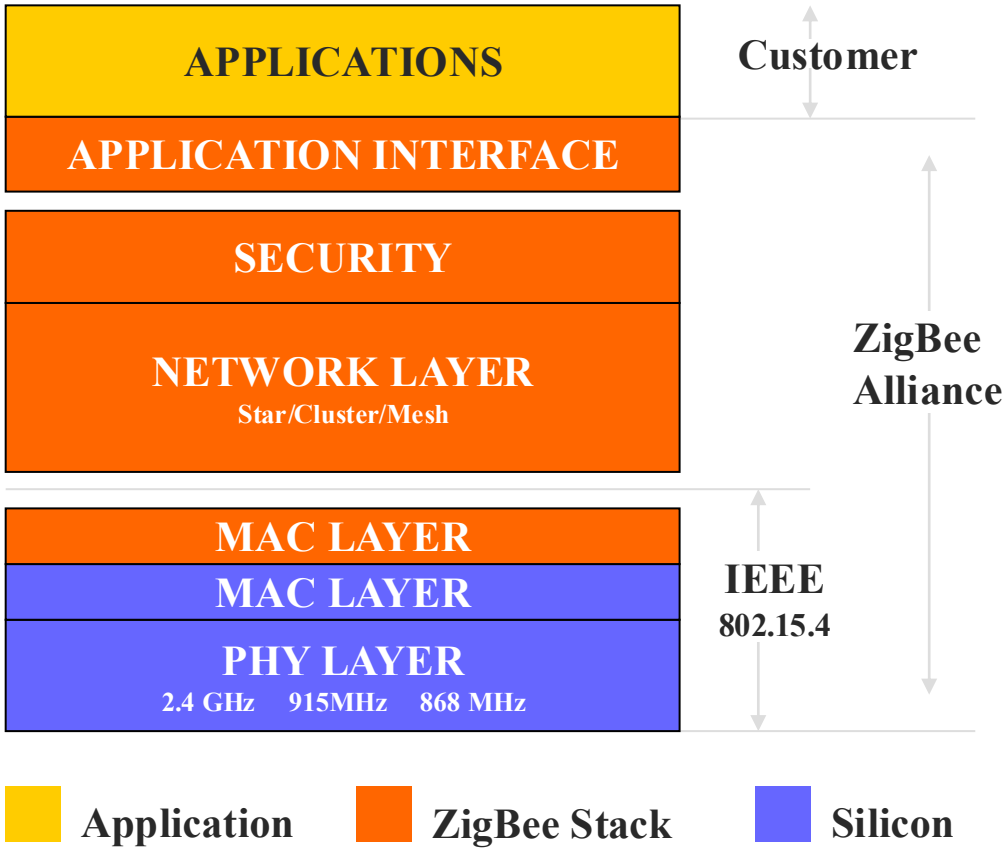
- Main (rough) difference:
  - THREAD extends IP to the Internet of Things
  - Zigbee doesn't
- Similarities in the lower layers







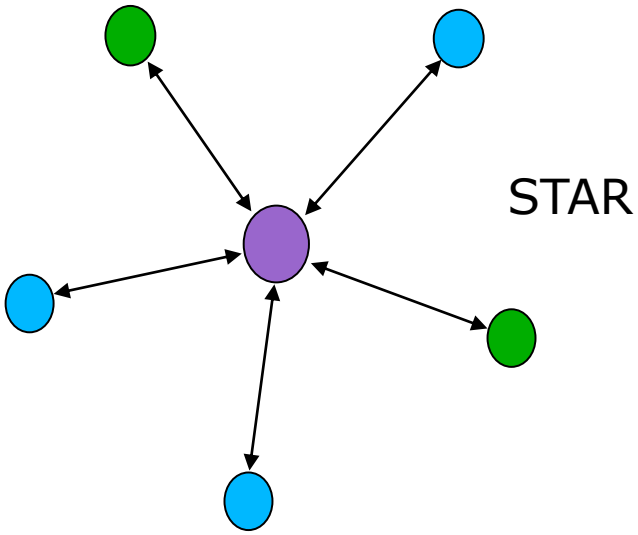
Zigbee: Communication Stack

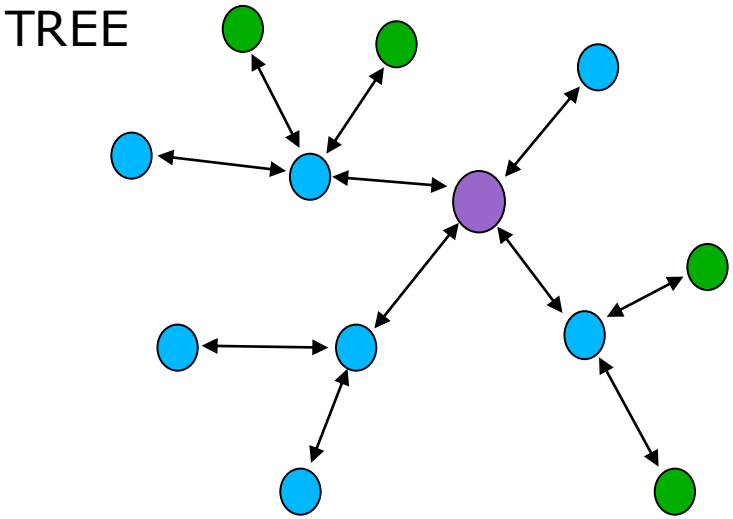


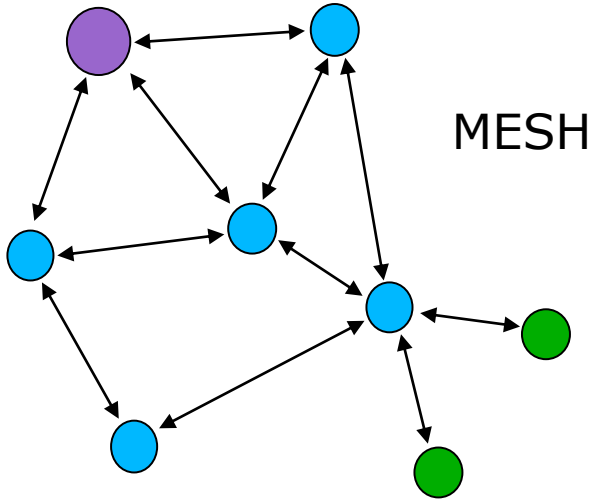
802.15.4: Types of Devices

- **Full Function Device (FFD):**
  - Can send beacons
  - Can communicate with other FFDs
  - Can route frames
  - Can act as PAN coordinator
  - Typically features power supply
- **Reduced Function Device (RFD):**
  - Cannot route frames
  - Cannot communicate with other RFDs
  - Can communicate with FFD
  - Runs typically on batteries
- **PAN Coordinator**
  - Is responsible of a Personal Area Network (PAN)
  - Manages PAN association/de-association

Supported Topology







PAN Coordinator

Full Function Device

Reduced Function Device

802.15.4: PHY

PHY (MHz)	Frequency band (MHz)	Spreading parameters		Data parameters		
		Chip rate (kchip/s)	Modulation	Bit rate (kb/s)	Symbol rate (ksymbol/s)	Symbols
868/915	868–868.6	300	BPSK	20	20	Binary
	902–928	600	BPSK	40	40	Binary
868/915 (optional)	868–868.6	400	ASK	250	12.5	20-bit PSSS
	902–928	1600	ASK	250	50	5-bit PSSS
868/915 (optional)	868–868.6	400	O-QPSK	100	25	16-ary Orthogonal
	902–928	1000	O-QPSK	250	62.5	16-ary Orthogonal
2450	2400–2483.5	2000	O-QPSK	250	62.5	16-ary Orthogonal

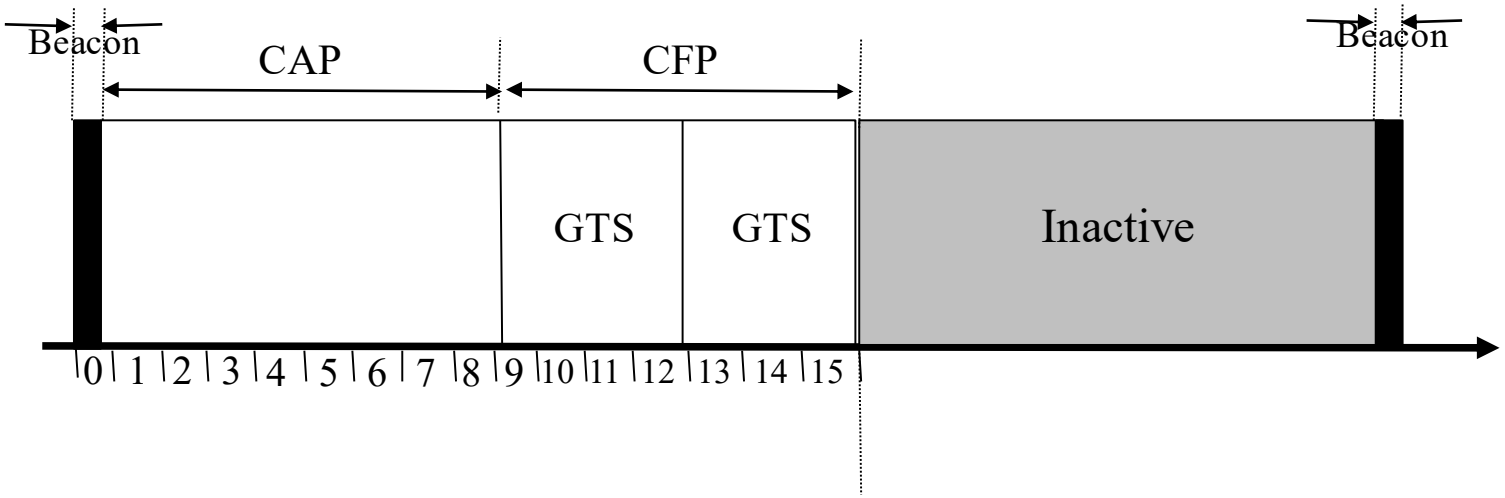
- Activation and deactivation of the radio transceiver
- Energy detection (ED) within the current channel
  - Detect energy level for each channel (used to implement scanning functionalities)
- Link quality indicator (LQI) for received packets
- Clear channel assessment (CCA)
  - Used to implement the carrier sense multiple access with collision avoidance (CSMA-CA)
- Channel frequency selection/Data transmission and reception

MAC: Functional Description

- A Mixture of Scheduled and Random Access
- Two operation modes are defined:
  - **Beacon Enabled**
    - PAN coordinator periodically transmits beacons
    - Usually adopted in star topologies
    - Slotted CSMA/CA + scheduled transmissions
  - **Non Beacon Enabled**
    - Uncoordinated access through unslotted CSMA/CA

Beacon Enabled: functional description

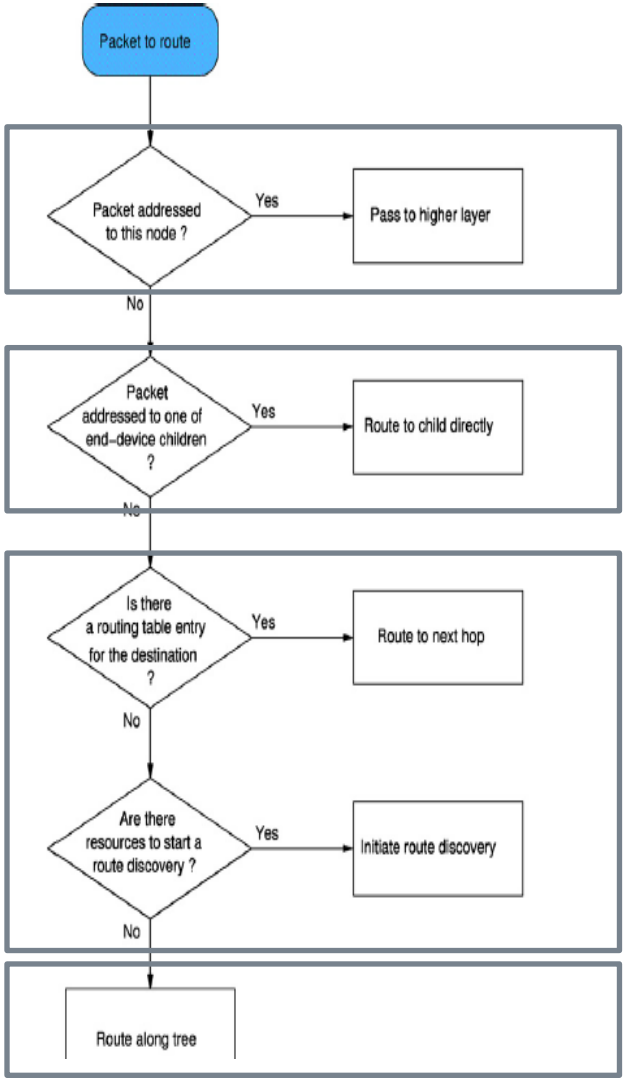
- **Beacon Enabled**



- Frame Length: from 15[ms] to 252[s] (15.38ms*2n where $0 \leq n \leq 14$)
- RFD associated to PAN coordinator are aligned with the frame structure
- Random access through CSMA/CA in CAP
- *Guranteed Time Slot* statically assigned by PAN coordinator through beacons

Zigbee Routing: overview

- Zigbee Specification (7/2005)
- Three Types of Devices:
  - *ZB Coordinator (FFD)*
  - *ZB Router (FFD)*
  - *ZB End-Device (RFD o FFD)*
- ZigBee Routing Integrates:
  - *Ad-hoc On-demand Distance Vector (AODV)*
  - *Cluster Tree Algorithm*



Local-destined packet

Destination One Hop Away

AODV routing

Fall-Back Tree-Based

Cluster Tree Algorithm: Tree Formation

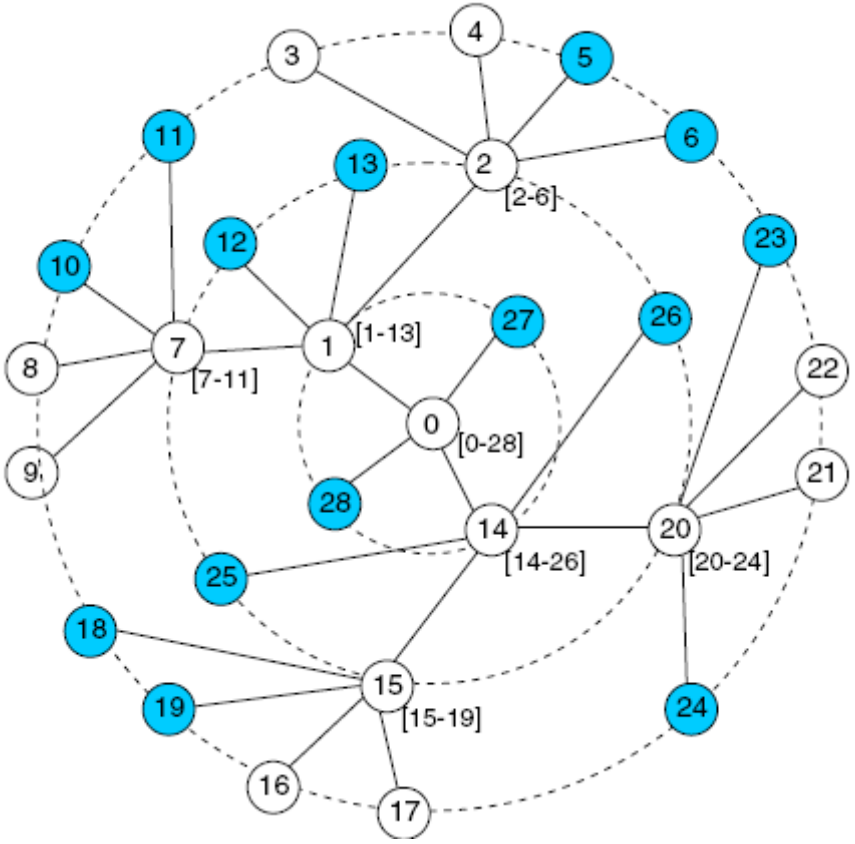
- A FFD kicks off the procedure:
  - It scans the available channels through the proper functionalities at the lower layers
  - Chooses a channel (e.g., the least interfered)
  - Sets the PAN identifier
  - Sets its own Network Address to 0 (Coordinator)
- Other devices may now associate to the coordinator through the lower-layer association procedures
- Associated devices may be:
  - ZB Router (only FFD): may let other devices to associate to the network
  - ZB End-Device
- Address Assignment (16 bits short addresses) is performed jointly with association
- Each parent device (PAN coordinator, ZB router) assigns groups of addresses to its children (other ZB routers, ZB end devices)

Cluster Tree Formation: principles

- On the basis of its depth in the tree, a newly joined router is assigned a range of consecutive addresses (16-bit integers).
- The first integer in the range becomes the node address while the rest will be available for assignment to its children (routers and end-devices).

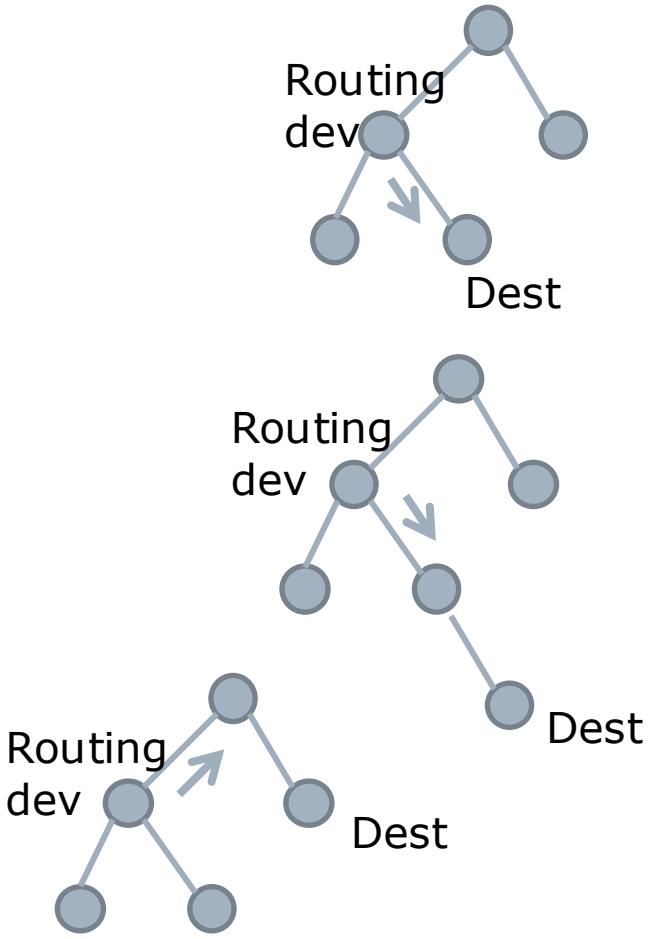
An Example

- PAN Coordinator can assign addresses in the range [0,28]



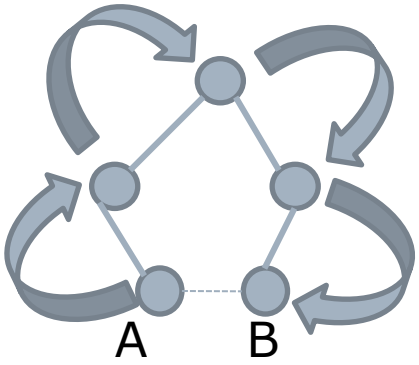
Tree-Based Routing: Principles

- Routing Along the Tree:
  - If destination address is one of children end devices:
    - route directly
  - Else if destination address belongs to one of children routers' addresses set:
    - send to corresponding children router
  - Else
    - Send to parent node



Routing Along the Tree: Shortcomings

- Routing may be not optimized
  - Route always along the tree
  - Routing is “quality-agnostic”
  - E.g.: A wants to send to B



AODV Routing

- A node willing to send to a destination broadcast a *Route Requests (RREQ)* message
  - aka shout ”where’s the destination”
- RREQ messages are flooded by receiving nodes
  - relay shouting
- When a node re-broadcasts a *Route Request*, it sets up a reverse path pointing towards the source
  - stores “who shouted at me”
- When the intended destination receives a Route Request, it replies by sending a Route Reply
  - Shouts back “It’s me”
- Route Reply travels along the reverse path set-up when Route Request is forwarded
  - Shouting travels back the same route

Routing Cost

- The cost for path *P* composed of *L-1* links is defined as:

$$C\{P\} = \sum_{i=1}^{L-1} C\{[D_i, D_{i+1}]\}$$

- ZigBee standards "suggests" the following form for the cost of the generic link *l*

$$C\{l\} = \begin{cases} 7, \\ \min\left(7, \text{round}\left(\frac{1}{p_l^4}\right)\right) \end{cases}$$

- P_l is the packet reception rate over link *l*

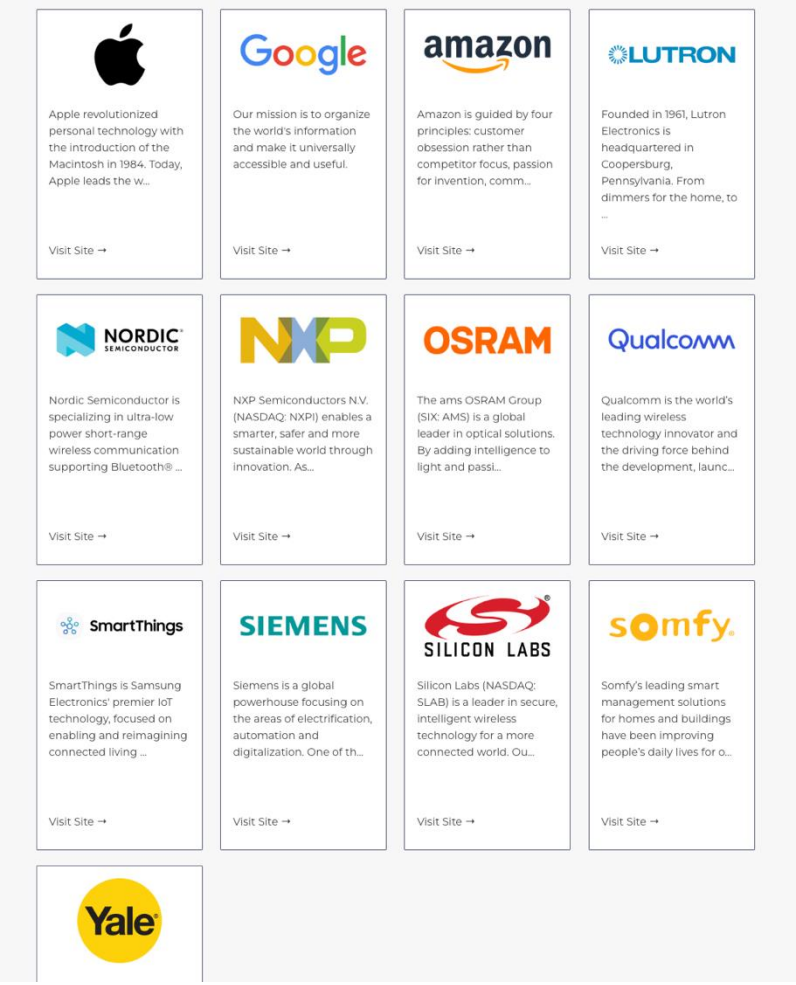
ZigBee Application Profiles

- Needs:
  - A common language for exchanging data
  - A well defined set of processing actions
  - Device interoperability across different manufacturers
  - Simplicity and reliability for the end users



- Profile Definition
  - A set of devices required in the application area
  - A set of clusters to implement the functionality
    - A set of attributes to represent device state
    - A set of commands to enable the communication
  - Specification of which clusters are required by which devices
  - Specific functional description for each device

THREAD https://www.threadgroup.org/

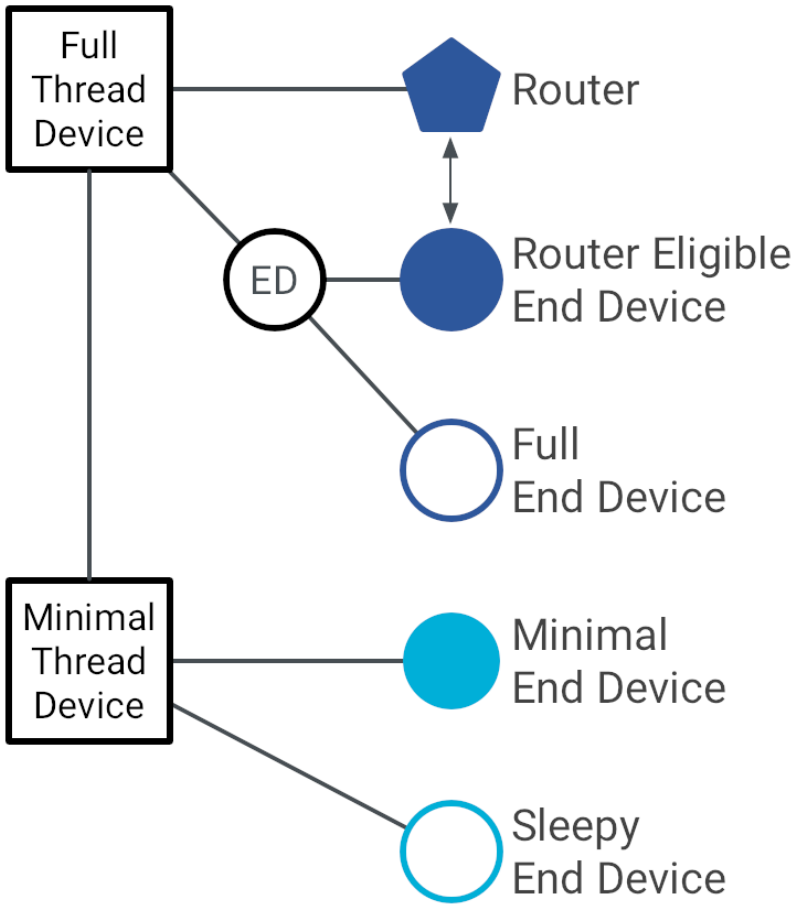


A low-power and low-latency **wireless mesh networking** protocol built using open and proven standards.

THREAD	Standard
Application Layer	
UDP + DTLS	RFC 768, RFC 6347, RFC 4279, RFC 4492v RFC 3315, 5007
Distance Vector Routing	RFC 1058, RFC 2080
6LowPAN (IPv6)	RFC 4944, RFC 4862, RFC 6775
IEEE 802.15.4 MAC (including MAC security)	IEEE 802.15.4 (2006)
IEEE 802.15.4 PHY	IEEE 802.15.4 (2006)

THREAD Devices – types and roles

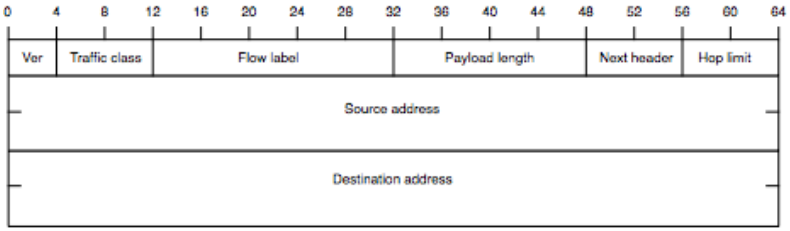
- Full Thread Device
  - ***Router***: route packets, always active,
  - ***Leader*** elected role of one Router, always active
  - ***Routing-eligible End Device*** (REED): can become router
  - ***Full End Device*** (FED)
- Minimal Thread Device
  - ***Minimal End Device*** (MED), always on
  - ***Sleepy End Device***, duty cycling
  - ***Synchronized Sleepy End Device***, duty cycling at scheduled time
- Boarder Router



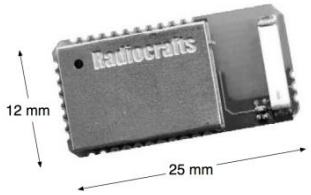
(Synchronized sleepy end device)

What is 6LoWPAN?

- An Adaptation Layer to fit IPv6 over Low-Power wireless Area Networks
- Defined by IETF standards
  - RFC 4919, 4944
  - draft-ietf-6lowpan-hc and -nd
  - draft-ietf-roll-rpl
- Efficient header compression
  - IPv6 base and extension headers, UDP header
- Fragmentation
  - 1280 byte IPv6 MTU -> 127 byte 802.15.4 frames

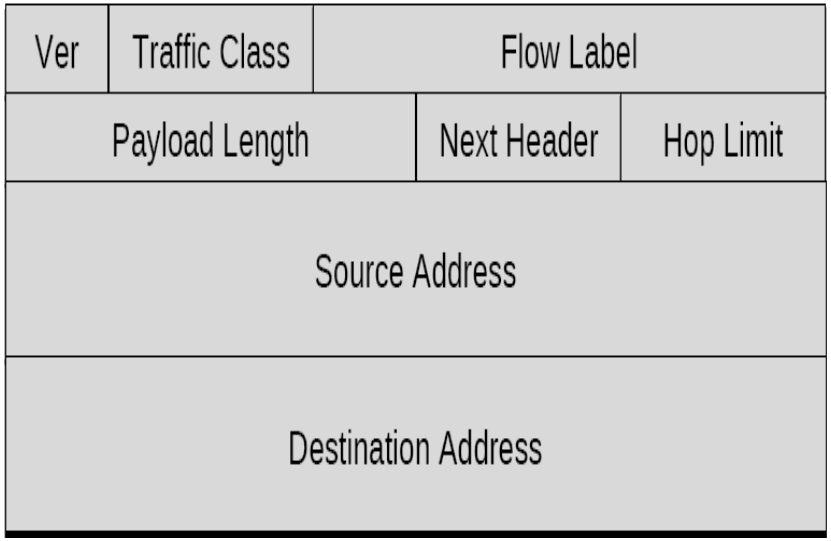


IPv6

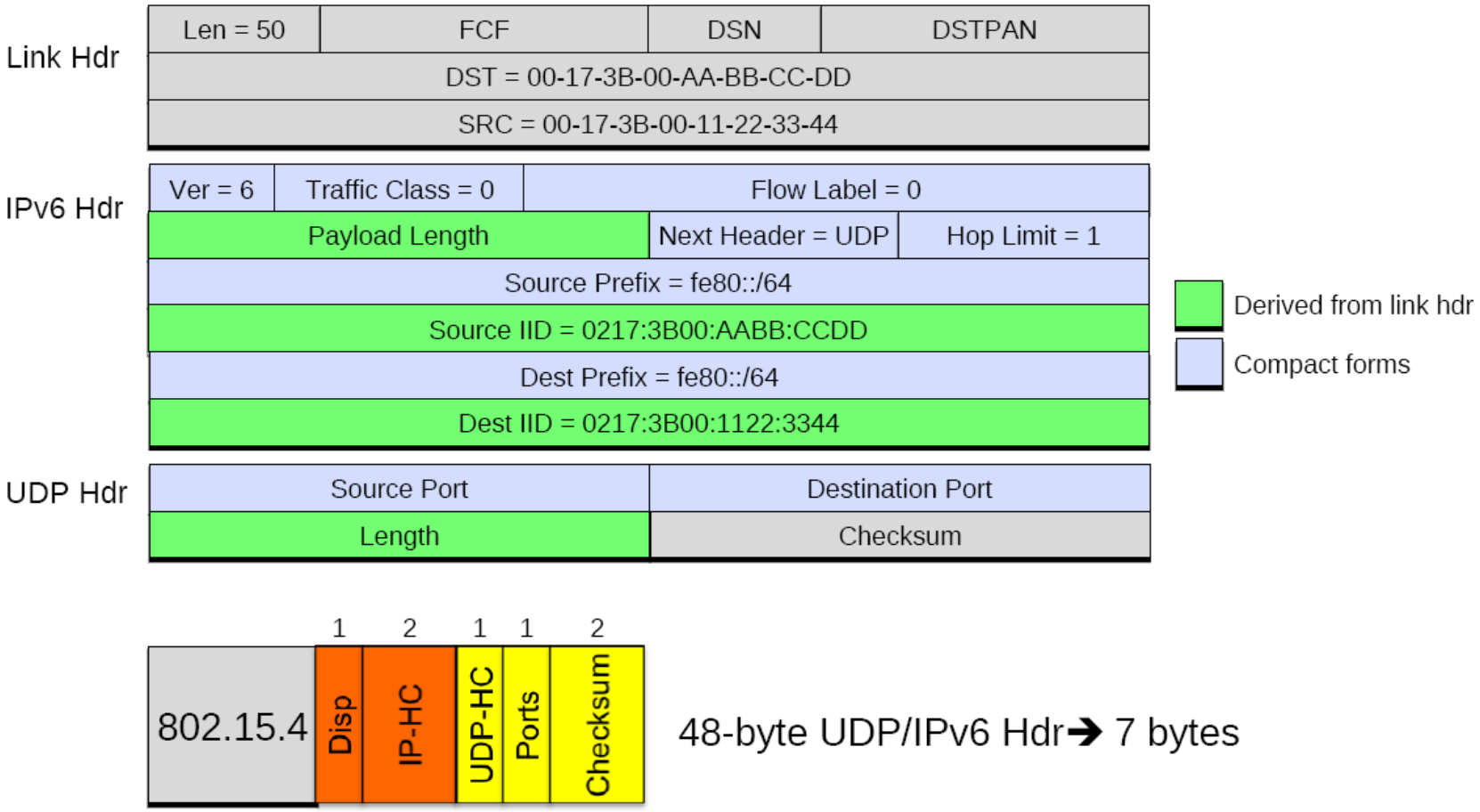


6LowPAN Header Compression at a Glance

- Stateless compression
- Flow-independent compression
- Simple tricks on IPv6/UDP header
  - Common values for header fields => compact forms
  - Version is always 6
  - Traffic Class and Flow Label are zero
  - Payload Length always derived from L2 header
  - Source and Destination Addresses can be elided (link-local) and/or compressed depending on the “context” of the transmission



Link Local


Link Hdr
Len = 50
FCF
DSN
DSTPAN
DST = 00-17-3B-00-AA-BB-CC-DD
SRC = 00-17-3B-00-11-22-33-44
IPv6 Hdr
Ver = 6
Traffic Class = 0
Flow Label = 0
Payload Length
Next Header = UDP
Hop Limit = 1
Source Prefix = fe80::/64
Source IID = 0217:3B00:AABB:CCDD
Dest Prefix = fe80::/64
Dest IID = 0217:3B00:1122:3344
UDP Hdr
Source Port
Destination Port
Length
Checksum
Derived from link hdr
Compact forms
1
2
1
1
2
802.15.4
Disp
IP-HC
UDP-HC
Ports
Checksum
48-byte UDP/IPv6 Hdr➔ 7 bytes

THREAD routing, Proactive, distance vector

- Each router has a routing data base
  - ***Route id set***: list of routers ID and SN
  - ***Link set***: list of current/recent neighbors

<l_router_id, l_link_marging, l_incoming_quality, l_outgoing_quality, l_age>

  - ***Route set***: routing table

<destination, next_hop, route_cost>

- Periodic signalling messages
  - Router ID, incoming_quality, outgoing_quality, route cost, sequence_number

Routing metrics

- Link margin and link quality

Link Margin	Quality	Link Cost
> 20dB	3	1
> 10dB	2	2
> 2dB	1	4
≤ 2dB	0	infinite

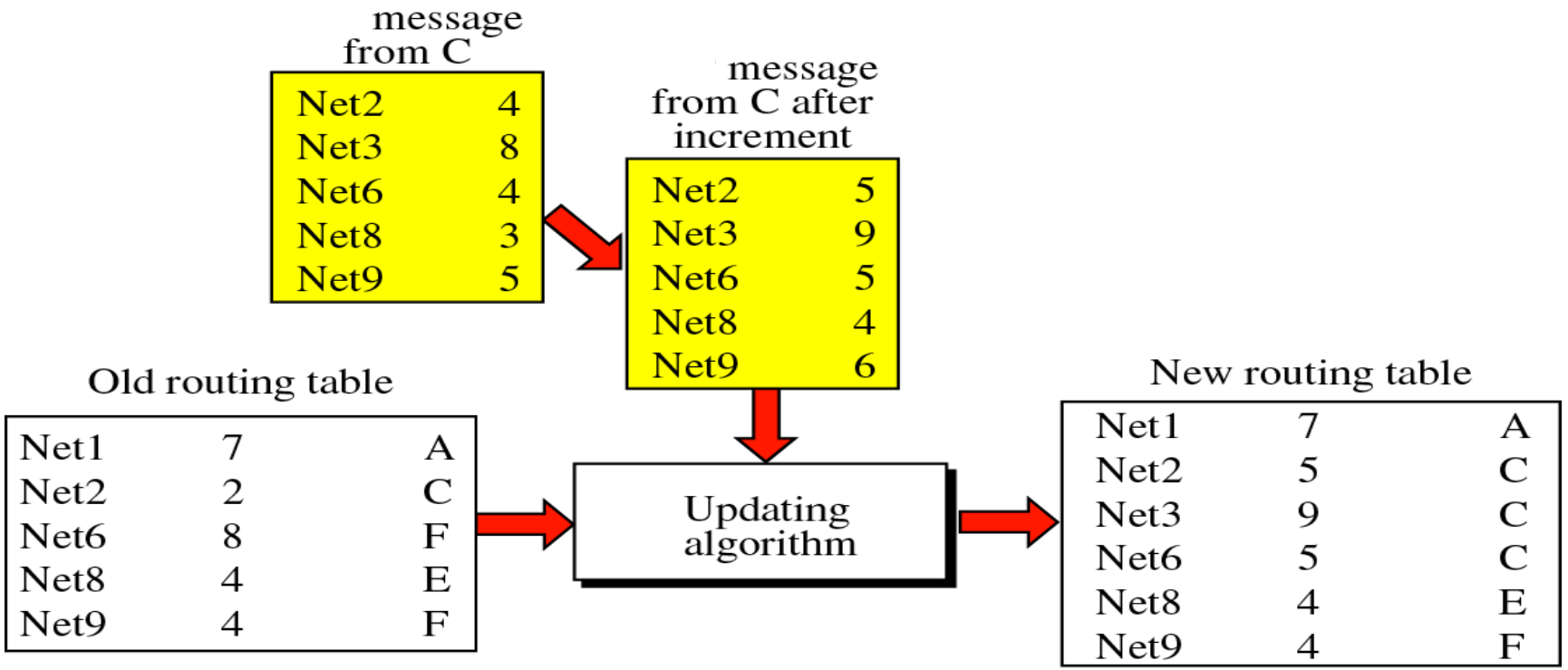
- Link margin is obtained through the exchange of periodic messages (64Route TLV advertisment)
- Link margin values are low-pass filtered
- Link quality is min(incoming_quality, outgoing:quality)

Distributed Bellman-Ford

Upon receiving a Route64 TLV (Distance Vector)

1. Update link quality
2. For each destination in *Route64 TLV*
   - If destination not in *route set*
     - Add destination/cost in *route set*
   - Otherwise
     - If next hop route set is equal to *Rout64 TLV* sender
       - Modify route set entry with new info in Route64 TLV
     - Otherwise
       - If cost *Rout64 TLV* is lower than the one store in route set
         - Modify route set entry with new info in Route64 TLV
3. Return

DV Example



Rules

Net1: No news, don't change
Net2: Same next hop, replace
Net3: A new router, add
Net6: Different next hop, new hop count smaller, replace
Net8: Different next hop, new hop count the same, don't change
Net9: Different next hop, new hop count larger, don't change

Bluetooth

Bluetooth vs. 802.15.1

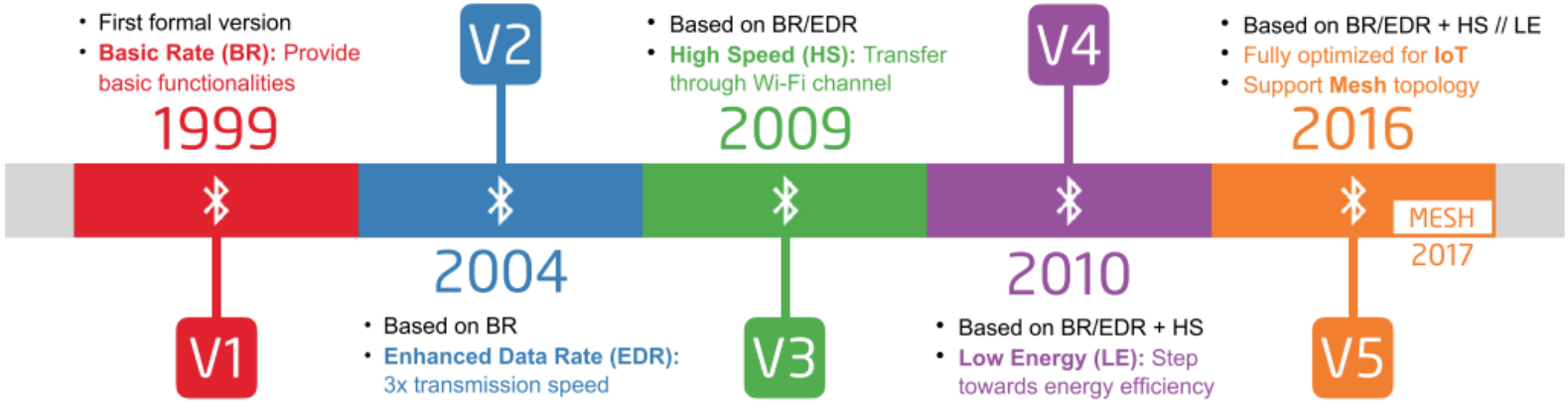
Bluetooth is an industrial specification for WPANs

The WG 802.15.1 adapted the industrial specifications of Bluetooth for the levels 1 and 2

’96-’97: Ericsson internal project

’98: Bluetooth SIG created (Ericsson, IBM, Intel, Toshiba, Nokia)

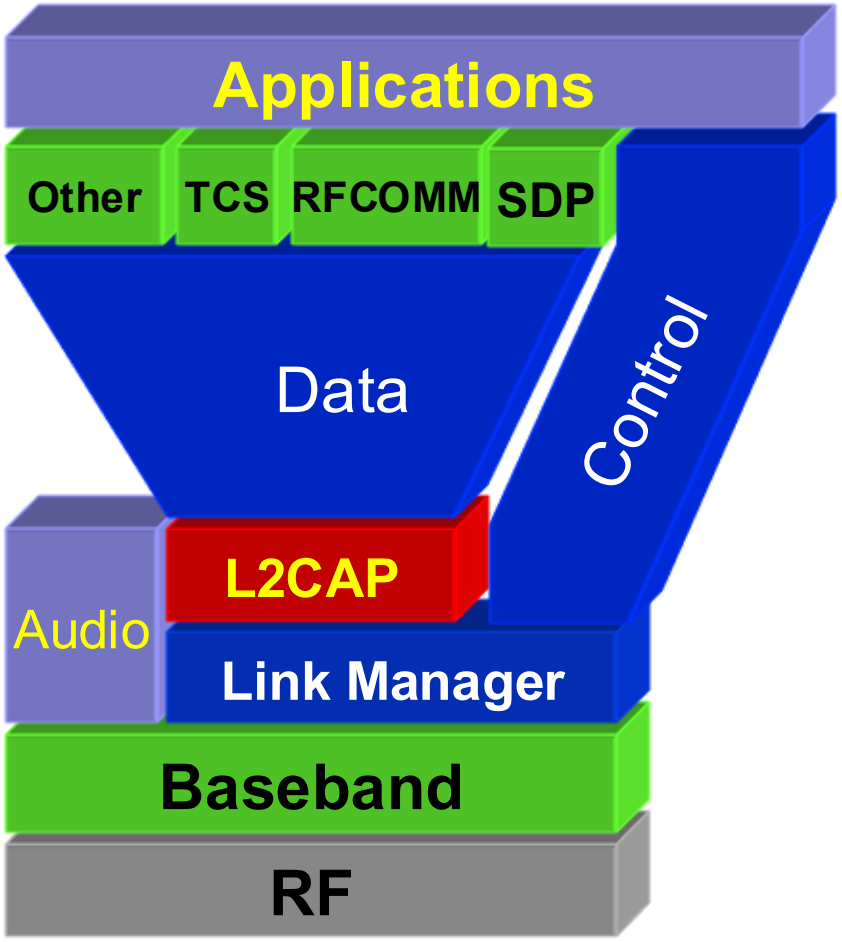
’99: new members join the SIG (3Com, Lucent Technologies, Microsoft, Motorola)



Bluetooth in Short

- There are essentially two strains of Bluetooth
  - the older legacy or classic strain that encompasses versions 1.0 through 3.0 (including BR/EDR)
  - the other strain is the low-energy Bluetooth that includes versions 4.0, 4.1, 4.2, 5.0, 5.1 and 5.2
- The low-energy version uses a different radio technology than the classic strain.
  - 79 @ 1 MHz channels (Bluetooth)
  - 40 @ 2 MHz channels (BLE)
- Most of the newer Bluetooth chips actually contain both types of radios

Protocol Architecture



- Original protocol architecture not compliant with IEEE 802 structure
- Later adapted by IEEE for 802.15.1 specifications
- RF + Baseband equivalent to PHY + MAC
- Control plane for the network and connections creation

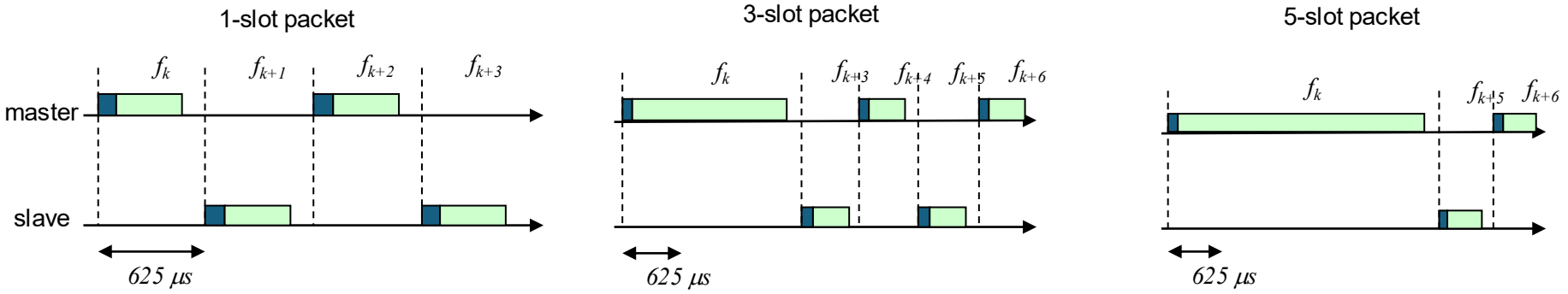
BR/EDR - Physical layer

- ISM band at 2.4 GHz
- 79 (23 in France and Japan) channels spaced of 1 MHz (2402-2480 MHz)
- Modulation G-FSK (1 Mb/s)
- Device classes

Class	Power (mW)	Power (dBm)	Range (approx)
Class 1	100 mW	20 dBm	~ 100 m
Class 2	2,5 mW	4 dBm	~ 10 m
Class 3	1 mW	0 dBm	~ 1 m

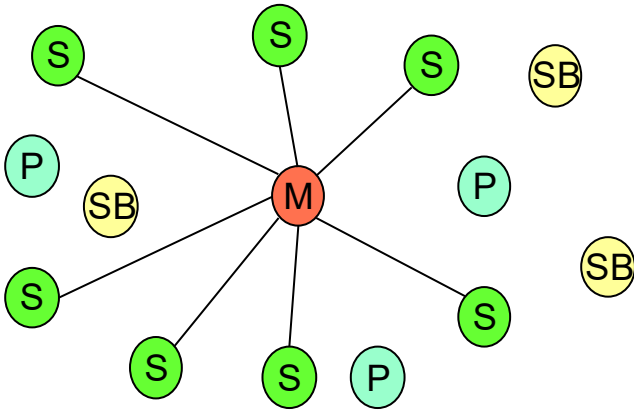
BR/EDR - Physical layer

- Frequency Hopping (FH) - 1600 hops/s (625 μs per hop)
- The FH sequence is pseudo random and determined by the clock and the address of the 'master' device that regulates the access to the channel
- The other devices are 'slaves' and follows the sequence f_k defined by the master



Piconet

- The simplest network architecture defined in Bluetooth is called piconet
- The piconet is an ad hoc network composed of 2 or more devices
- A device acts as master and the other as slaves
- Communication can take place only between master and slave and not directly between slaves
- Up to 7 slaves can be active in a piconet
- The others can be in
  - Stand-by (not part of the piconet)
  - Parked (part of the piconet but not active, up to a maximum of 256 devices)

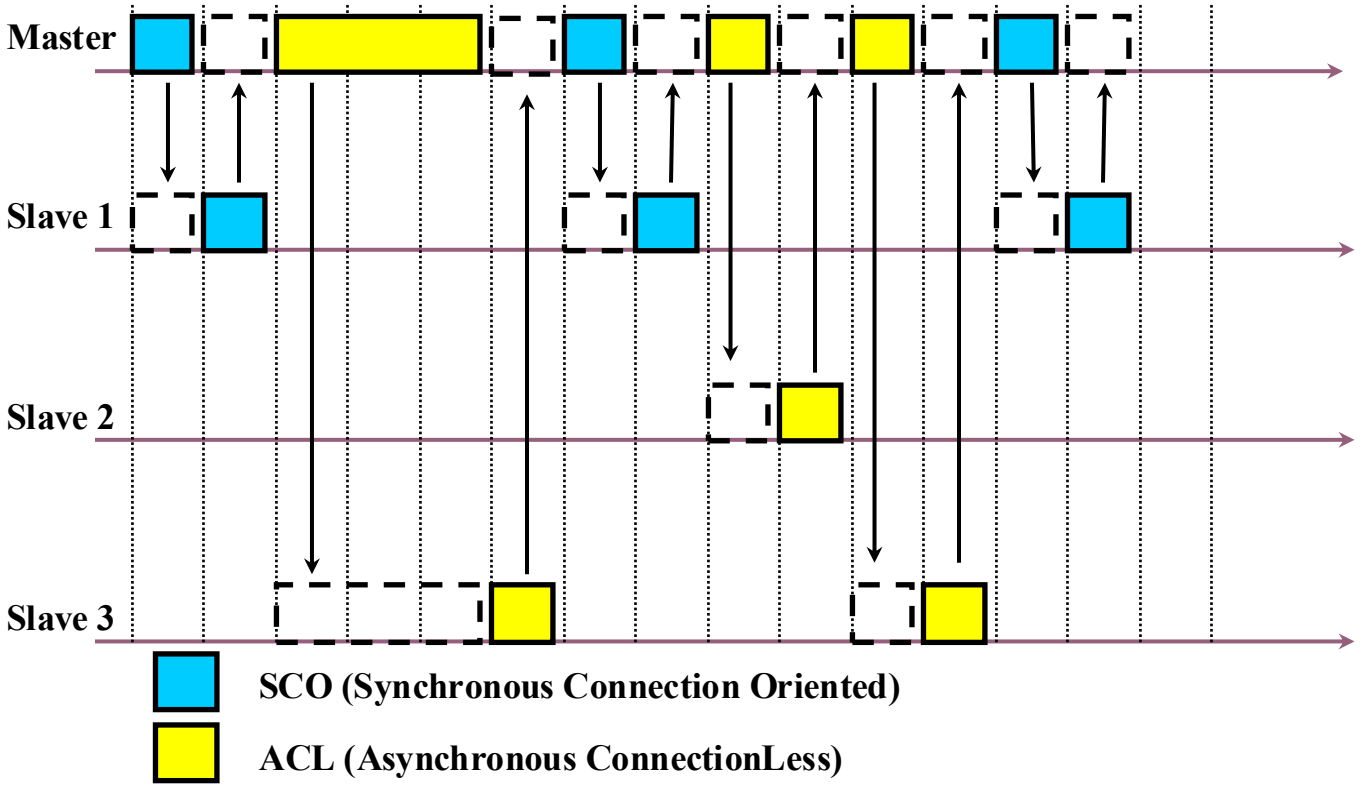


- Addresses
  - MAC address of 48 bits
  - AMA (Active Member Address) 3 bits
  - PMA (Parked Member Address) 8 bits

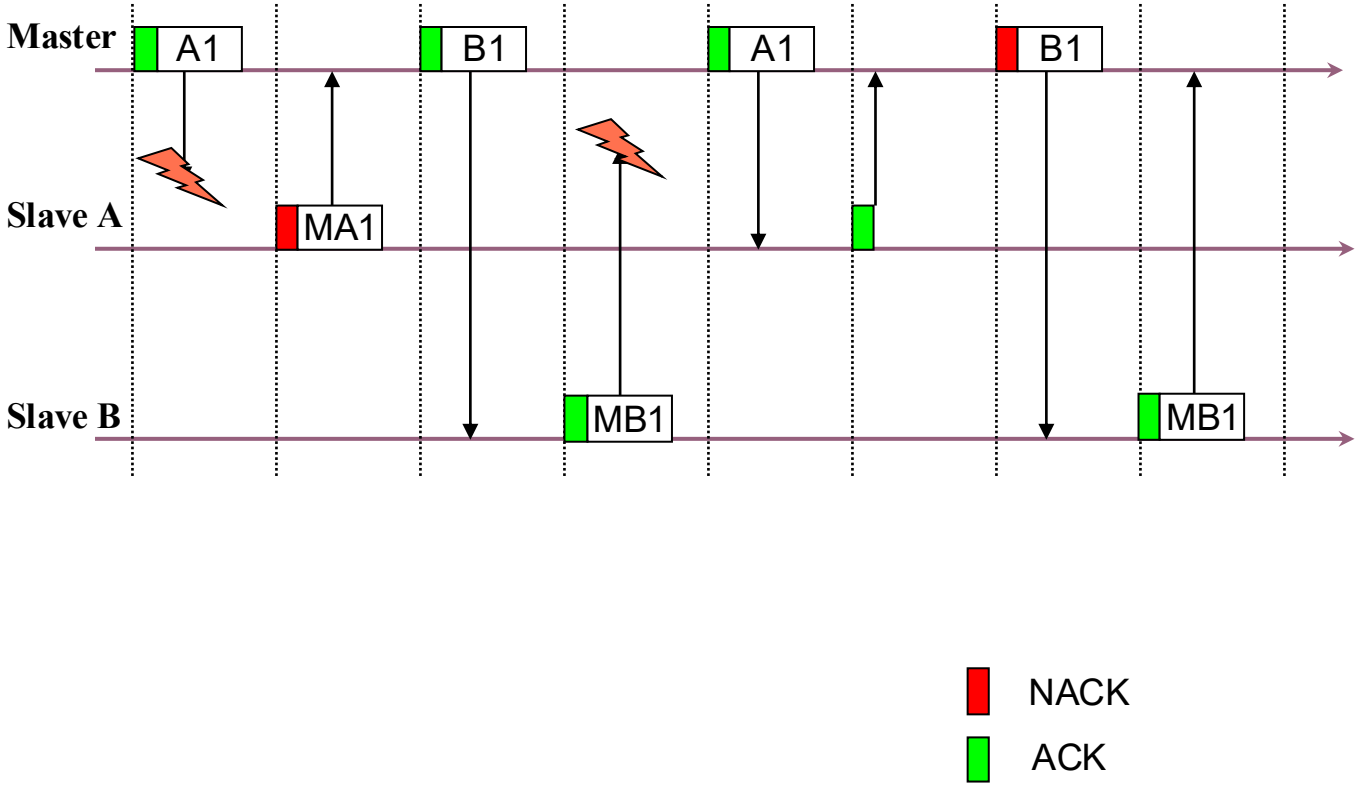
Types of connections

- Bluetooth considers two types of connections
- SCO (Synchronous Connection Oriented)
  - Fixed rate bi-directional connection (circuit)
  - FEC for improving quality
  - Rate of 64 Kbit/s
- ACL (Asynchronous ConnectionLess)
  - Packet switched connection shared between master and active slaves based on a polling access scheme
  - Several options for packet formats and physical layer codes (1, 3, 5 slots)
  - Rate up to 433.9 Kbit/s symmetric (using 5-slot packets in both directions) and 723.2/57.6 Kbit/s asymmetric (using 5-slot packets in one direction and 1-slot packets in the other)

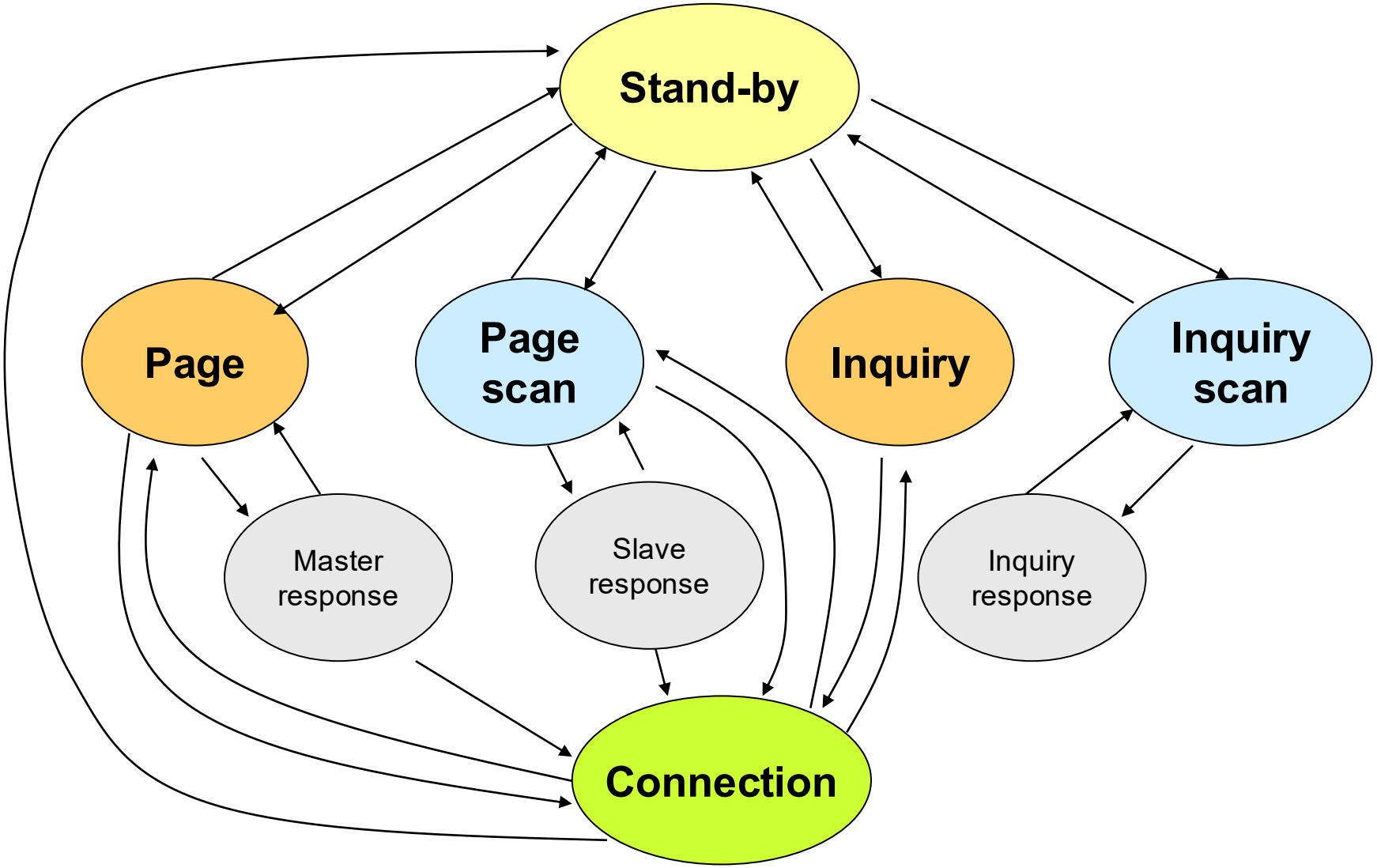
Multiple Access



Link controller: ARQ



Link controller: states



Profiles

- Basic operation modes characteristic of different applications
- Used to guaranteed interoperability at application layer

- Generic Access Profile
- Service Discovery Application Profile
- Cordless Telephony Profile
- Intercom Profile
- Serial Port Profile
- Headset Profile
- Dial-up Networking Profile
- Fax Profile
- LAN Access Profile
- Generic Object Exchange Profile
- Object Push Profile
- File Transfer Profile
- Synchronization Profile
- Advanced Audio Distribution
- PAN
- Audio Video Remote Control
- Basic Printing
- Basic Imaging
- Extended Service Discovery
- Generic Audio Video Distribution
- Hands Free
- Hardcopy Cable Replacement

Profiles



Specifications

Beats Wireless™ can make and receive calls hands-free with your compatible mobile phone and listen to music from a compatible music player, but they should be within 10 metres (33 feet) of each other.

The headset is compliant with the Bluetooth® Specification 2.1 + EDR and supports the following profiles: Headset Profile (HSP) V 1.2, Hands-Free Profile (HFP) V 1.5, Advanced Audio Distribution Profile (A2DP) V 1.0 and Audio Video Remote Control Profile (AVRCP) V 1.0. Input rating 5V DC, 0.5A.

PJ-763MFi

The following profiles are supported:

- SPP (Serial Port Profile)
- OPP (Object Push Profile)
- BIP (Basic Imaging Profile)
- HCRP (Hard Copy Cable Replacement Profile)
- iAP (iPod Accessory Protocol)



Bluetooth Low Energy (BLE)

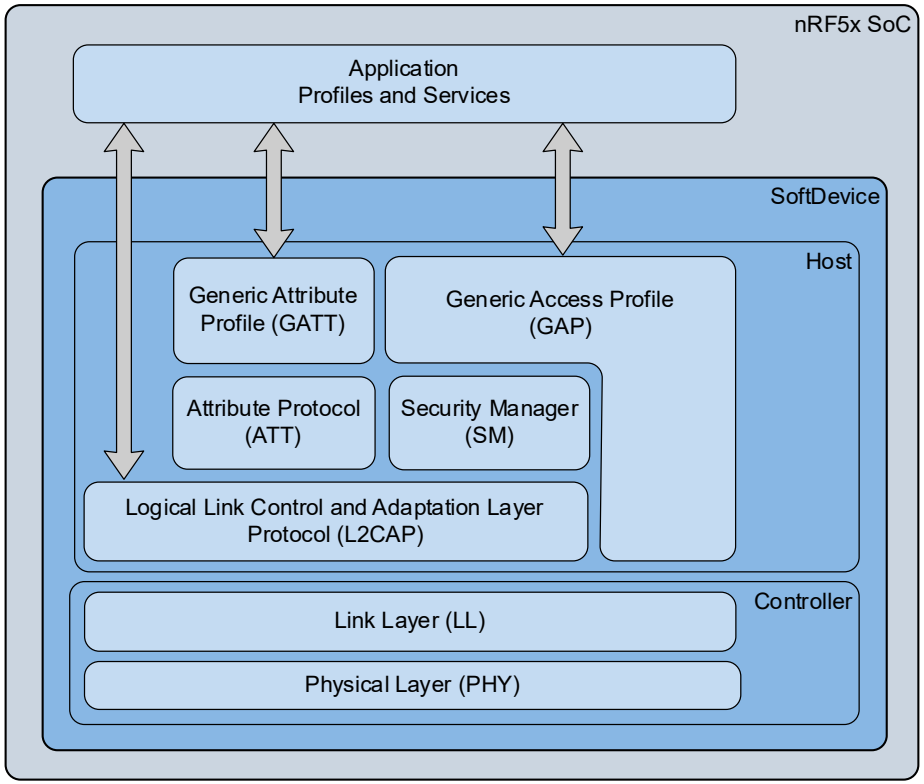
- BLE vs Bluetooth BR/EDR
  - Not replacing, but sitting alongside Bluetooth BR/EDR; **BLE and Bluetooth are not compatible**
  - Designed to be energy efficient, tailored to IoT application
  - Service/device discovery faster than Bluetooth BR/EDR (3ms vs 100ms)
  - Shorter packets/lower complexity

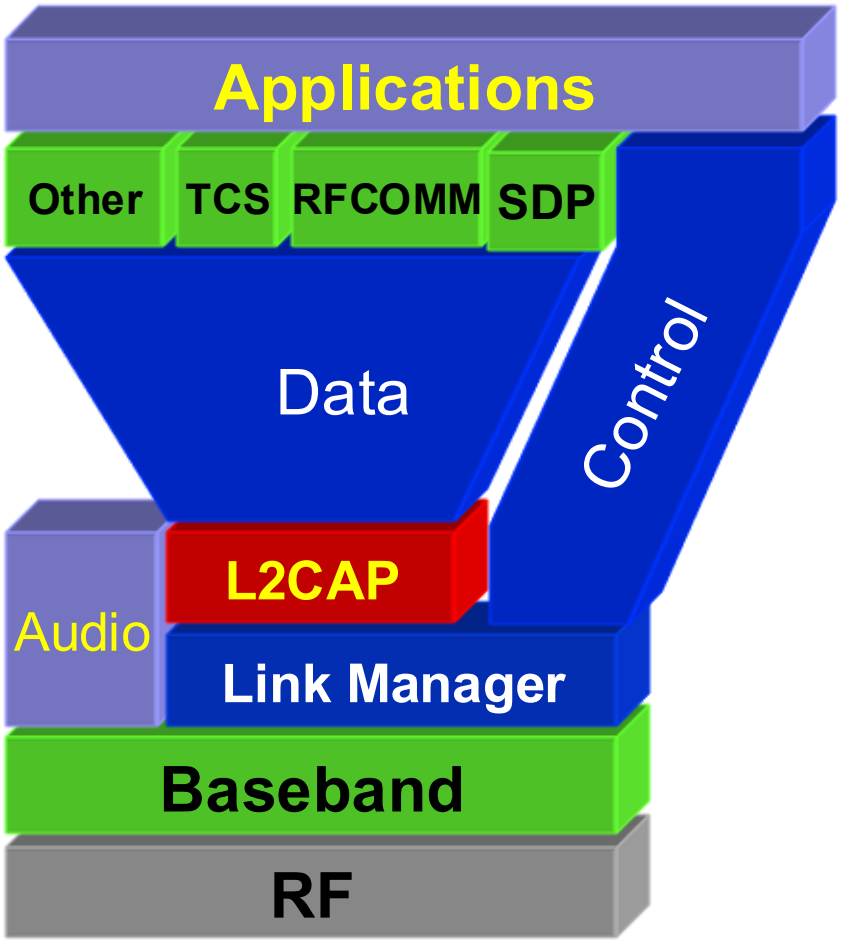
Table 1: Bluetooth BR/EDR vs. Bluetooth BLE

	Bluetooth BR/EDR	BLE
Channel scheme	79 channels	40 channels
Max output power	100 mW	10 mW
Latency	100 ms	6 ms
Time to send data	100 ms	3 ms
Raw data rate	1 Mb/s; 2-3 Mb/s	1 Mb/s
Throughput	0.7 Mb/s; 2.1 Mb/s	0.27 Mb/s
Max range	~100 m	~100 m

BLE vs Bluetooth: a different stack

- “Simpler” than Bluetooth





BLE Physical Layer

- Same ISM band used in BR/EDR, but:
  - 40 2MHz channels
    - 3 channels for advertisement/broadcast
    - 37 for data transfer via adaptive frequency hopping
  - GFSK modulation
  - Three PHY types:
    - LE 1M (1Mbps), no FEC, mandatory
    - LE 2M (2Mbps), no FEC, optional
    - LE Coded (0.5/0.125Mbps), FEC, optional. Used for extending range 2x or 4x

Central/Peripheral nodes

- Similar concept to master/slave
- Peripheral nodes transmit advertisements periodically (20ms – 10s) on 3 specific channels
- Centrals scan the adv channels for advertisements
  - scan window: for how long to scan
  - scan interval: how often to scan

BLE AFH

- BLE uses Adaptive Frequency Hopping on the 37 data channels
- Central node selects:
  - Channels to use (avoiding the ones with interference). Channel map is shared with peripheral node.
  - Hopping algorithm
    - Algorithm #1: sequential selection with a fixed hop interval passed during connection
    - Algorithm #2: semi-random sequence calculated from address/internal counters

ATtribute Protocol-ATT
A common ground to design applications

Client-Server information exchange protocol for BLE

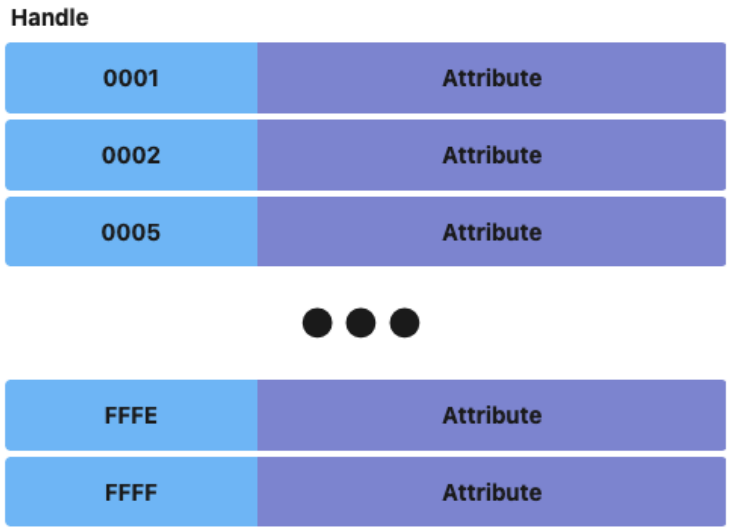
standard way to store the data at the server

standard way to access the data

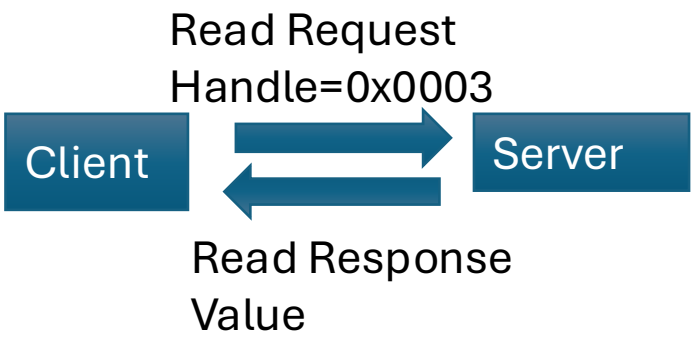
- **Data is structured in Attributes:**
  - **Handle** - 16-bit unique attribute identifier (does not change during a single connection)
  - **Attribute Type (UUID)** - 2-byte (16-byte) globally unique type/meaning of the specific data stored in the attribute (two types of UUIDs: Service UUID and Characteristic UUID)
  - **Attribute Value** - actual data being stored
  - **Attribute Permissions** - specifies privacy/security methods, readable/writable

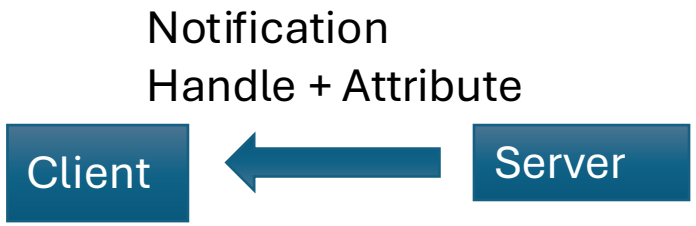


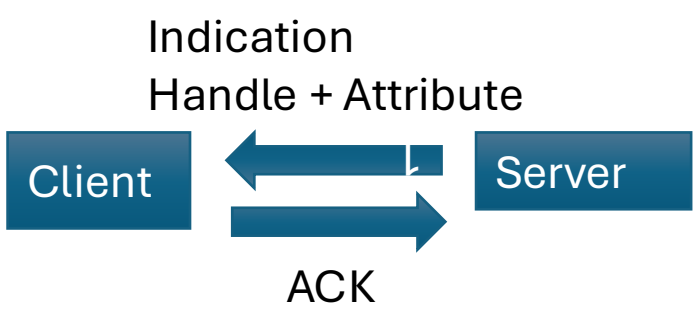
Attributes Access mode examples



Server's side database







Heart Rate Profile

- *Heart Rate Service* (HRS - UUID 0x180D): The primary service exposing sensor data.
  - *Heart Rate Measurement* (UUID 0x2A37): A mandatory characteristic that notifies the collector of heart rate (in bpm), energy expended, and R-R intervals.
  - *Body Sensor Location* (UUID 0x2A38): An optional characteristic that tells the client where the sensor is worn (e.g., finger, wrist, chest).
  - *Heart Rate Control Point* (UUID 0x2A39): An optional characteristic used to reset the energy expended value.

Example: Hearth Rate Profile

Characteristic	UUID	Property	Data Sent
Heart Rate Measurement	0x2A37	Notify	BPM (uint8/uint16), R-R intervals
Body Sensor Location	0x2A38	Read	Location enum (e.g., Other, Chest, Wrist)
Heart Rate Control Point	0x2A39	Write	Reset energy expended (0x01)

HRP Workflow

